Phase Equilibrium (相平衡)

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Course Requirements (本章课程要求)

- **Master:** Concepts of phase, number of components and degrees of freedom; physical significance of the phase rule and its application in phase diagrams; the Clausius-Clapeyron equation together with its applications and calculations for one-component systems; the lever rule and its application in phase diagrams.
- **Be familiar with:** Significance and applications of eutectic-system phase diagrams; p-x and T-x diagrams for binary liquid systems, characteristics of azeotropic systems; principles of distillation and rectification; composition representation of ternary systems; phase diagrams of partially-miscible ternary-liquid systems and their applications in extraction.
- **Understand:** Methods of constructing cooling curves from phase diagrams or drawing simple phase diagrams from cooling curves; principle and operation of ebulliometer; phase diagrams of aqueous-salt systems and their basic applications.

1. The Phase Rule (相律)

1.1 Phase and Number of Phases (相和相数)

■ Phase (相)

A phase is a homogeneous portion of a system that has uniform physical and chemical properties. There exist distinct interfaces between different phases, which can be separated by mechanical means. A phase may be either continuous or dispersed.

系统中，具有相同物理性质和化学性质完全均匀部分，称为相。相与相之间有明显的界面，可以用机械的方法使其分开。一个相可以是连续的，也可以是分散的。

1. The Phase Rule (相律)

1.1 Phase and Number of Phases (相和相数)

■ Number of phases (相数)

The number of phases present in a system is called the number of phases, denoted by the symbol Φ. The existence of a phase is independent of its quantity or volume.

系统中所具有相的数目，称为相数。以符号Φ表示。相的存在与其数量的多少以及体积的大小无关。

1. The Phase Rule (相律)

1.1 Phase and Number of Phases (相和相数)

■ Points to note when discussing the number of phases:

① The same system can have different phases and different numbers of phases under different conditions. For example, water can exist as one-phase, two-phase or three-phase equilibrium coexistence under different conditions.

② All gases in a system can mix at the molecular level. Therefore, there is only one gas phase (mixed gases constitute a single phase).

① 同一系统，在不同条件下可以有不同的相和相数。例如水在不同条件下可以是一相、两相、三相平衡共存。

② 系统中，无论有多少种气体，都可以达到分子程度的混合，因此只有一个气相（气体混合一个相）

1. The Phase Rule (相律)

1.1 Phase and Number of Phases (相和相数)

■ Points to note when discussing the number of phases:

③ When different liquids are mixed, the system may form one phase or two phases depending on their mutual solubilities (The number of liquid layers equals the number of liquid phases).

④ For mixtures of different solids, no matter how finely ground or thoroughly mixed the solids are, there will be as many phases as there are kinds of solids, provided molecular-level dispersion is not achieved. If the solids form a solid solution with molecular-level dispersion, the system becomes one single phase. For a pure solid with multiple crystal forms, there are as many phases as there are crystal modifications.

③ 不同液体混合时，依其相互溶解度不同，可以是一相、两相(几个液层几个相）；

④ 不同固体混合，无论这些固体研磨得多么细小，混合得多么均匀，只要未达到分子程度的分散，则有几种固体就有几相。若这些固体形成了分子程度分散的固溶体，则为一相。一种纯固体若有不同的晶形，则有几种晶形就有几相。

1. The Phase Rule (相律)

1.1 Phase and Number of Phases (相和相数)

■ Questions: Determine the number of phases in each of the following systems.

① Several pieces of ice in water, together with water-vapor; ② Mixture of iron powder and coal; ③ Two-liquid-phase system of water and oil; ④ Water and oil after vigorous shaking and stirring; ⑤ Several solid pieces of silica together with one piece of calcium oxide; ⑥ A large sodium chloride crystal mixed with fine sodium-chloride powder; ⑦ Mixed system of $CO_2(g)$, $N_2(g)$ and $H_2(g)$.

■ 小问题：请问下列系统相数是多少

①水中有几块冰，还有水蒸汽；②铁粉和煤的混合物；③水和油的双液系统；④ 水和油剧烈震荡、搅拌；⑤ 几块二氧化硅固体和一块氧化钙固体；⑥ 大块氯化钠晶体和微小氯化钠粉末混合；⑦$CO_2(g)$、$N_2(g)$和$H_2(g)$混合系统。

1. The Phase Rule (相律)

1.2 Number of Chemical Species and Components (物种数和组分数)

■ Number of chemical species (物种数)

The number of distinct chemical substances contained in a system, denoted by symbol S. These substances may be compounds or elementary substances. They can be isolated separately and exist stably for a long time. The same substance in different states of aggregation counts as only one chemical species. The value of species number S may vary according to the way the problem is considered.

系统中所含化学物质的种数称为物种数。用符号S表示。它们可以是化合物或单质。能够单独分离出来，并长期稳定存在。处于不同聚集状态的同一物质，只能算一个物种。物种数随着人们考虑问题的方式不同而不同。

1. The Phase Rule (相律)

1.2 Number of Chemical Species and Components (物种数和组分数)

■ **Example 1 for Number of chemical species (物种数):**

A mixed aqueous system of NaOH and H_3PO_4. If chemical reactions are not considered: S=3 (NaOH, H_3PO_4 and H_2O). If chemical reactions are taken into account: S=6 (NaOH, H_3PO_4, H_2O, Na_3PO_4, Na_2HPO_4 and NaH_2PO_4).

NaOH和H_3PO_4的水溶液混合系统，如不考虑化学反应的发生：S=3（NaOH，H_3PO_4和H_2O)，如考虑化学反应的发生：S=6（NaOH，H_3PO_4，H_2O，Na_3PO_4，Na_2HPO_4和NaH_2PO_4)

1. The Phase Rule (相律)

1.2 Number of Chemical Species and Components (物种数和组分数)

■ **Example 2 for Number of chemical species (物种数):**

Aqueous NaCl solution system. If ionization is not considered: S=2 (NaCl and H_2O). If ionization is considered: S=6 (NaCl, H_2O, H^+,OH^-, Na^+ and Cl^-).

Thought question: Can H^+,OH^-, Na^+ and Cl^- could be isolated individually? Can they exist stably for a long time?

NaCl水溶液系统。若不考虑电离：S=2（NaCl和H_2O）若考虑电离：S=6（NaCl, H_2O, H^+,OH^-, Na^+和Cl^- ）思考：试问H^+, OH^-, Na^+和Cl^-能单独分离出来吗？能长期稳定存在吗？

1. The Phase Rule (相律)

1.2 Number of Chemical Species and Components (物种数和组分数)

■ **Number of components (组分数):**

The number of the minimum species required to express the composition of each phase in a phase-equilibrium system is defined as the number of components, denoted by the symbol K. The number of chemical species and the number of components are two distinct concepts. Their values may be identical or different.

表示相平衡系统中各相组成所需的最少物种数称为组分数。用符号K表示。物种数和组分数是两个不同的概念，其数值有时相同，有时不同。

1. The Phase Rule (相律)

1.2 Number of Chemical Species and Components (物种数和组分数)

■ **Number of components (组分数):**

When no chemical reactions occur in the system: S = K

When the system contains R independent chemical reaction equations and R' independent concentration-restriction conditions: K = S-R-R'

当系统中不存在化学反应时：S = K;

当系统中存在R个独立的化学反应式数和 R′个独立的浓度限制条件时：K = S–R–R′。

1. The Phase Rule (相律)

1.2 Number of Chemical Species and Components (物种数和组分数)

■ **Number of components (组分数):**

Several explanations for the concentration-restriction term R' are given below:

- **If the system contains only HI(g) and reaches chemical equilibrium upon heating: the reaction is 2HI (g) = H_2 (g) + I_2(g); S=3, R=1, R′ =1 ($n_{H_2} : n_{I_2} = 1:1$), K=1**
- **若系统中只有HI (g)，加热达化学平衡时：因有2HI (g) = H_2 (g) + I_2(g); S=3, R=1, R′ =1 ($n_{H_2} : n_{I_2} = 1:1$), K=1**

1. The Phase Rule (相律)

1.2 Number of Chemical Species and Components (物种数和组分数)

■ **Number of components (组分数):**

Several explanations for the concentration-restriction term R' are given below:

- However, the concentration-restriction condition R′ must apply within the same phase. Consider the system: $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$. At chemical equilibrium, R'=0. Even though the molar ratio $n_{CO2}:n_{CaO}$ = 1:1, these two substances do not exist in the same phase.
- 但是浓度限制条件R′要求必须在同一相中。系统为: $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$ 达化学平衡时，R′= 0，虽然$n_{CO2}:n_{CaO}$= 1:1，但它们不在同一相中。

1. The Phase Rule (相律)

1.2 Number of Chemical Species and Components (物种数和组分数)

■ Number of components (组分数):

Several explanations for the concentration-restriction term R' are given below:

- For a system consisting of HI(g) together with small amounts of H_2(g) and I_2(g), chemical equilibrium is established upon heating via the reaction: 2HI (g) = H_2 (g) + I_2 (g). Although this reaction produces H_2 and I_2 in a stoichiometric ratio, small quantities of H_2(g) and I_2(g) are already present in the original system. Therefore, there is no concentration-restriction condition, and R′= 0.
- 若HI (g)，少量的H_2 (g)和I_2 (g)构成的系统，加热达化学平衡时，有：2HI (g) == H_2 (g) + I_2 (g)，虽然反应中 有 $n_{H_2}:n_{I_2}=1:1$ 但是系统中原来就有少量的H_2 (g)和I_2 (g) 。所以：$n_{H_2总}:n_{I_2总}\neq 1:1$，就没有浓度限制条件了，R′= 0。

1. The Phase Rule (相律)

1.2 Number of Chemical Species and Components (物种数和组分数)

■ Number of components (组分数):

Several explanations for the concentration-restriction term R' are given below:

- **An evacuated vessel contains solid $NH_4HCO_3(s)$. Equilibrium is achieved upon heating, with the reaction: $NH_4HCO_3\ (s)=NH_3(g)\ +CO_2(g)\ +H_2O\ (g)$. The amounts-of-substance of any two components are constrained by the third. Hence, R′=2.**
- **一抽空的容器中放有$NH_4HCO_3(s)$，加热达平衡。发生下列反应：$NH_4HCO_3\ (s)=NH_3(g)\ +CO_2(g)\ +H_2O\ (g)$，因有：其中两个组分的物质的量受到另外一个的限制所以：R′=2。**

1. The Phase Rule (相律)

1.2 Number of Chemical Species and Components (物种数和组分数)

■ Number of components (组分数):

Several explanations for the concentration-restriction term R' are given below:

- In the system of water, benzene and benzoic acid, the concentration of benzoic acid in water is specified. If the concentration of one component is specified, then R' = 1; if the concentrations of two components are specified, then R' = 2.
- 水、苯和苯甲酸系统中，指定水中苯甲酸的浓度。指定某个组分的浓度，则R′=1；若指定两个组分的浓度，则R′=2。

1. The Phase Rule (相律)

1.2 Number of Chemical Species and Components (物种数和组分数)

Question: A system consists of arbitrary amounts $NH_4Cl(S)$, $NH_3(g)$, $HCl(g)$. Determine the number of species and the number of components under each of the following conditions:

a) At low temperature, no chemical reaction occurs; b) At high temperature, chemical equilibrium is established; c) Evacuate the system, then allow it to reach chemical equilibrium.

小问题：任意量的 $NH_4Cl(S)$, $NH_3(g)$、$HCl(g)$混合为系统。在下述条件下，物种数和组分数各为多少?

a) 低温，无任何反应；b) 高温，达化学平衡；c) 抽真空后，再达平衡。

1. The Phase Rule (相律)

1.3 Degrees of freedom (自由度)

■ **Concept of degrees of freedom (自由度的概念):**

Under the condition that no old phases disappear and no new phases are generated, the intensive properties that can vary freely within a certain range are defined as the degrees of freedom (T, P, concentrations) of the equilibrium system under given conditions, denoted by the symbol f.

在不引起旧相消失和新相生成的条件下，可在一定范围内自由变动的强度性质，称为平衡系统在该条件下的自由度（T、P、浓度），用符号f 表示。

1. The Phase Rule (相律)

1.3 Degrees of freedom (自由度)

■ Example of degrees of freedom (自由度的举例说明):

Experience tells us that when only water vapor exists in a system, the temperature and pressure can be independently varied within a certain range without the disappearance of this gas phase or the formation of new phases. In this case, the degrees of freedom f = 2. When the system consists of liquid water coexisting with water vapor at equilibrium, only one intensive property, either temperature or pressure, can be independently changed to keep both the liquid-water and gas phases from vanishing, that is, f = 1.

例如，由经验得知，当系统中只有水蒸气存在时，若要这个气相不消失又不产生新相，可以在一定范围内独立地改变温度和压力，此时，自由度f = 2。当系统是由液态水与其平衡的水蒸气构成时，若要保持水及气两相都不消失，能独立改变的强度性质只有一个，温度或压力，即f = 1。

1. The Phase Rule (相律)

1.3 Degrees of freedom (自由度)

■ **Example of degrees of freedom (自由度的举例说明):**

When the system is composed of liquid water, ice and water vapor in equilibrium, neither temperature nor pressure can be changed if all three phases are to be preserved, f = 0 (0.0098℃; 610Pa).

当系统是由液态水、冰与其平衡的水蒸汽构成时，若要保持三相都不消失，温度和压力都不能变，即f=0。(0.0098℃；610Pa)。

1. The Phase Rule (相律)

1.4 Phase rule (相律)

■ Concept of phase rule (相律的概念):

In a phase-equilibrium system, the rule describing the relationship among the number of components, the number of phases, the degrees of freedom and external influencing factors (such as T and P) can be expressed as: $f = K-\varphi+2$, where K denotes the number of components; φ stands for the number of phases; and the constant 2 refers to the two external variables, temperature and pressure.

在相平衡系统中，表示相数、组分数、自由度数及外界影响因素（如T、P）之间关系的规律，可以证明：$f = k-\varphi+2$，式中 K 是组分数；φ是相数；2 是指温度、压力两个外界影响因素。

1. The Phase Rule (相律)

1.4 Phase rule (相律)

■ Concept of phase rule (相律的概念):

This equation can be transformed under constant temperature or constant pressure as: f = k–φ+1; under both constant temperature and constant pressure, it becomes: f = k–φ. The proof is given below: the number of degrees of freedom equals the number of independently variable variables, which is equal to the total number of variables minus the number of independent relational expressions among the variables.

该式在定温或定压下可变为：f= k –φ+1；在定温、定压下变为：f= k –φ；证明如下：自由度数=能独立改变的变量数=总变量数–变量间独立的关系式数。

1. The Phase Rule (相律)

1.4 Phase rule (相律)

Question: When solid ammonium chloride is heated, part of it decomposes into ammonia and hydrogen-chloride gas. Analyze the degrees of freedom of the system once equilibrium is established. (1) Heat the solid after evacuating the container; (2) Heat the solid when arbitrary amounts of ammonia gas and hydrogen-chloride gas are already present in the container.

小问题：加热固体NH_4Cl，一部分分解为NH_3和HCl气体，当系统建立平衡时，试分析系统的自由度是多少？(1) 容器抽真空后开始加热；(2) 加热时容器中已有任意量的NH_3 (g)和HCl (g)。

1. The Phase Rule (相律)

1.4 Phase rule (相律)

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小问题：加热固体NH_4Cl，一部分分解为NH_3和HCl气体，当系统建立平衡时，试分析系统的自由度是多少？(1) 容器抽真空后开始加热；(2) 加热时容器中已有任意量的NH_3 (g)和HCl (g)。

答案： (1) 此系统S=3 (NH_4Cl；NH_3；HCl)；R=1，NH_4Cl (s) = NH_3(g)+HCl (g)；R'=1，nNH_3:nHCl=1:1，则组分数: K=S–R–R'=3–1–1=1；相数：φ=2；根据相律f=K–φ+2=1–2+2=1。

(2) 此时系统物种数仍为3，仍有一化学平衡关系，R=1，但没有浓度限制条件，R'=0；组分数：K=S–R–R'=3–1–0=2；相数：φ=2；自由度f=K–φ+2=2–2+2=2。

2. Phase Equilibrium of One-Component System (单组分系统相平衡)

2.1 Two-Phase Equilibrium in One-Component Systems-The Clausius Clapeyron Equation (单组分系统的两相平衡—克劳修斯–克拉贝龙方程)

For a system with the number of components equal to 1, it contains only one pure substance, and its phase-rule expression is: $f=k-\varphi+2=1-\varphi+2=3-\varphi$; When f = 0, the number of phases reaches its maximum: $\varphi=3$; When φ = 1, the degree of freedom reaches its maximum: f = 2.

组分数为1的系统只含有一种纯物质，其相律表达式为：$f=k-\varphi+2=1-\varphi+2=3-\varphi$

当f=0时，相数最多φ= 3；当φ=1时，自由度最多 f = 2

2. Phase Equilibrium of One-Component System (单组分系统相平衡)

2.1 Two-Phase Equilibrium in One-Component Systems-The Clausius Clapeyron Equation (单组分系统的两相平衡—克劳修斯–克拉贝龙方程)

When φ=2, f=1. Only one of temperature and pressure can be varied independently, and the other changes accordingly. There exists a certain functional relationship between them. Examples include the relationship between saturated vapor pressure of liquid and temperature, the relationship between saturated vapor pressure of solid and temperature, as well as the relationship between external pressure and the melting point of solid.

当φ=2时，f=1，温度和压力中只有一个是可以独立改变的，另一个则随之变化，两者间存在着某种函数关系。例如液体的饱和蒸气压与温度的关系；固体的饱和蒸气压与温度的关系 以及外压与固体的熔点的关系等。

2. Phase Equilibrium of One-Component System (单组分系统相平衡)

2.1 Two-Phase Equilibrium in One-Component Systems-The Clausius Clapeyron Equation (单组分系统的两相平衡—克劳修斯–克拉贝龙方程)

It can be proved that for a one-component system at two-phase equilibrium:

$$\frac{dP}{dT} = \frac{\Delta H_m}{T\Delta V_m}$$

the above equation is known as the Clapeyron equation. It describes the rate of change of pressure with temperature under two-phase equilibrium, where △Vm denotes the molar volume difference between the two phases, and △Hm is the molar enthalpy of phase transition.

可以证明，单组分系统两 相平衡时：

$$\frac{dP}{dT} = \frac{\Delta H_m}{T\Delta V_m}$$

上式称为克拉贝龙方程，它表明两相平衡时压力随温度的变化率，式中△Vm是两相的摩尔体积差，△Hm是摩尔相变热。

2. Phase Equilibrium of One-Component System (单组分系统相平衡)

2.1 Two-Phase Equilibrium in One-Component Systems-The Clausius Clapeyron Equation (单组分系统的两相平衡—克劳修斯–克拉贝龙方程)

The Clapeyron equation is the specific form of f=1, namely T=f(P) or P=f(T), for two-phase equilibrium in a one-component system derived from thermodynamics.

克拉贝龙方程就是由热力学推导出的单组分系统的两相平衡时，f =1： T= f (P)或 P=f (T)的具体形式。

2. Phase Equilibrium of One-Component System (单组分系统相平衡)

2.1 Two-Phase Equilibrium in One-Component Systems-The Clausius Clapeyron Equation (单组分系统的两相平衡—克劳修斯–克拉贝龙方程)

Detailed analysis is presented below.

① Liquid-gas two-phase equilibrium. If V_m^l is neglected and the vapor is treated as an ideal gas, the Clapeyron equation becomes:

$$\frac{dP}{dT}=\frac{\Delta H_m}{RT^2}\cdot P\text{或}\frac{d\ln p}{dT}=\frac{\Delta H_m}{RT^2}$$

The formula above is the differential form of the Clausius-Clapeyron equation.

下面进行详细分析

① 液-气两相平衡，若忽略 V_m^l ,并将蒸气视为理想气，克拉贝龙方程变为：

$$\frac{dP}{dT}=\frac{\Delta H_m}{RT^2}\cdot P\text{或}\frac{d\ln p}{dT}=\frac{\Delta H_m}{RT^2}$$

上式称为克劳修斯－克拉贝龙方程的微分式。

2. Phase Equilibrium of One-Component System (单组分系统相平衡)

2.1 Two-Phase Equilibrium in One-Component Systems-The Clausius Clapeyron Equation (单组分系统的两相平衡—克劳修斯–克拉贝龙方程)

When the temperature variation range is small, △Hm can be regarded as a constant. Performing indefinite integration: $$\ln p = -\frac{\Delta H_m}{R}\cdot\frac{1}{T} + C$$

C is the integration constant. It can be seen that there exists a linear relationship between $\ln p \sim \frac{1}{T}$ with a slope of $-\frac{\Delta H_m}{R}\cdot$ This formula can be used to calculate the molar enthalpy of vaporization of a pure liquid.

当温度范围变化不大时，△Hm可视为常数。进行不定积分：$$\ln p = -\frac{\Delta H_m}{R}\cdot\frac{1}{T} + C$$

C为积分常数，可看出$\ln p \sim \frac{1}{T}$是直线关系，斜率为$-\frac{\Delta H_m}{R}\cdot$由此式可计算纯液体的摩尔汽化热。

2. Phase Equilibrium of One-Component System (单组分系统相平衡)

2.1 Two-Phase Equilibrium in One-Component Systems-The Clausius Clapeyron Equation (单组分系统的两相平衡—克劳修斯–克拉贝龙方程)

Definite integration:

$$\ln\frac{p_2}{p_1}=\frac{\Delta H_m}{R}\left(\frac{1}{T_1}-\frac{1}{T_2}\right)$$

P1 is the saturated vapor pressure of the pure liquid at temperature T1; P2 is the saturated vapor pressure of the pure liquid at temperature T2; △Hm is the molar enthalpy of vaporization of the pure liquid.

定积分：

$$\ln\frac{p_2}{p_1}=\frac{\Delta H_m}{R}\left(\frac{1}{T_1}-\frac{1}{T_2}\right)$$

式中：

P1为T1时纯液体的饱和蒸汽压；P2为T2时纯液体的饱和蒸汽压；△Hm为纯液体的摩尔汽化热。

2. Phase Equilibrium of One-Component System (单组分系统相平衡)

2.1 Two-Phase Equilibrium in One-Component Systems-The Clausius Clapeyron Equation (单组分系统的两相平衡—克劳修斯–克拉贝龙方程)

This equation can be applied to calculate the saturated vapor pressure of a pure liquid at different temperatures, its boiling point under different external pressures, and its molar enthalpy of vaporization.

$$\ln\frac{p_2}{p_1}=\frac{\Delta H_m}{R}\left(\frac{1}{T_1}-\frac{1}{T_2}\right)$$

应用该式可计算纯液体在不同温度时的饱和蒸气压；不同外压下的沸点以及摩尔汽化热。

2. Phase Equilibrium of One-Component System (单组分系统相平衡)

2.1 Two-Phase Equilibrium in One-Component Systems-The Clausius Clapeyron Equation (单组分系统的两相平衡—克劳修斯–克拉贝龙方程)

Example: A steam boiler can withstand a pressure of 10×101325 Pa. At what heating temperature will the boiler become hazardous? Given that the enthalpy of vaporization of water is 2255 $J \cdot g^{-1}$.

Solution: It is known that the normal boiling point of water is 373.2 K at a pressure of 101325 Pa, and $\triangle Hm = 2255 \times 18$ J.

例：有一水蒸气锅炉，能耐10×101325 Pa的压力，问此锅炉加热到什么温度有危险？已知水的汽化热为2255 $J \cdot g^{-1}$。

解：已知水的正常沸点为373.2K，此时压力为101325 Pa，$\triangle Hm = 2255 \times 18$ J

2. Phase Equilibrium of One-Component System (单组分系统相平衡)

2.1 Two-Phase Equilibrium in One-Component Systems-The Clausius Clapeyron Equation (单组分系统的两相平衡—克劳修斯–克拉贝龙方程)

According to the Clausius-Clapeyron equation:

$$\ln\frac{p_2}{p_1}=\frac{\Delta H_m(T_2-T_1)}{RT_2T_1}$$

Substitute T1=373.2 K, P1=101325 Pa, P2=10 ×101325 Pa, △Hm=2255 ×18 J into the equation, and we obtain T2=470.3 K.

根据克-克方程:

$$\ln\frac{p_2}{p_1}=\frac{\Delta H_m(T_2-T_1)}{RT_2T_1}$$

将T1=373.2K，P1=101325Pa，P2=10×101325Pa，△Hm =2255×18J；代入：得T2= 470.3K

2. Phase Equilibrium of One-Component System (单组分系统相平衡)

2.1 Two-Phase Equilibrium in One-Component Systems-The Clausius Clapeyron Equation (单组分系统的两相平衡—克劳修斯–克拉贝龙方程)

Trouton Rule

This empirical rule can be used for approximate estimation when enthalpy-of-vaporization data for liquids are unavailable. For normal liquids (non-polar, non-associated liquids),

it states that:

$$\frac{\Delta_{vap}H_m}{T_b} \approx 88J \cdot K^{-1} \cdot mol^{-1}$$

特鲁顿规则：当缺乏液体的汽化焓数据时，可用此经验性规则进行近似估计。

例如对正常液体（非极性、非缔合性液体）来说有：

$$\frac{\Delta_{vap}H_m}{T_b} \approx 88J \cdot K^{-1} \cdot mol^{-1}$$

2. Phase Equilibrium of One-Component System (单组分系统相平衡)

2.1 Two-Phase Equilibrium in One-Component Systems-The Clausius Clapeyron Equation (单组分系统的两相平衡—克劳修斯–克拉贝龙方程)

② Solid-Gas Two-Phase Equilibrium

Similarly, Vm >> Vm(s). Treating the vapor as an ideal gas and taking the molar enthalpy of sublimation as a constant yields the same relational formula as that for liquid-gas equilibrium. These expressions describe the relationship between the saturated vapor pressure of a pure solid and temperature, where △Hm denotes the molar enthalpy of phase transition.

② 固-气两相平衡

同样Vm(g) >> Vm(s)；把蒸气看成理想气体及将摩尔升华热视为常数可以得到与液-气平衡相同的关系式。它们表明纯固体的饱和蒸气压与温度的关系，△Hm是摩尔相变热。

2. Phase Equilibrium of One-Component System (单组分系统相平衡)

2.1 Two-Phase Equilibrium in One-Component Systems-The Clausius Clapeyron Equation (单组分系统的两相平衡—克劳修斯–克拉贝龙方程)

③ Solid-Liquid Two-Phase Equilibrium

The temperature for solid-liquid two-phase equilibrium of a pure substance is its melting point. Since Vm (s) and V(l) do not differ significantly and neither can be neglected, the Clapeyron equation is written as:

$$\frac{dT}{dP}=\frac{T\Delta V_m}{\Delta H_m}$$

where △Hm is the molar enthalpy of fusion. This formula expresses the relationship between the melting point of a pure solid and the equilibrium external pressure.

③ 固-液两相平衡：纯物质固 - 液两相平衡的温度是熔点，由于Vm（s）与Vm(l)相差不大，不能忽略，将克拉贝龙方程写为 $\frac{dT}{dP}=\frac{T\Delta V_m}{\Delta H_m}$

式中△Hm为摩尔熔化热，公式表示纯固体的熔点与平衡外压的关系。

2. Phase Equilibrium of One-Component System (单组分系统相平衡)

2.1 Two-Phase Equilibrium in One-Component Systems-The Clausius Clapeyron Equation (单组分系统的两相平衡—克劳修斯–克拉贝龙方程)

③ Solid-Liquid Two-Phase Equilibrium

When the pressure change is small or for approximate calculations, △Hm and △Vm can be regarded as constants, and definite integration is performed on the above equation:

$$\ln\frac{T_2}{T_1}=\frac{\Delta V_m}{\Delta H_m}(p_2-p_1)$$

③ 固-液两相平衡:

当压力变化不大或近似计算时，可将△Hm和△Vm看成常数，将上式进行定积分:

$$\ln\frac{T_2}{T_1}=\frac{\Delta V_m}{\Delta H_m}(p_2-p_1)$$

2. Phase Equilibrium of One-Component System (单组分系统相平衡)

2.2 Phase Diagram of Water (水的相图)

- Phase diagram: A geometric plot drawn from experimental results, which describes the relationships between equilibrium phases in a system and their corresponding conditions (T, P, X).
- Phase diagrams can be classified according to the number of components: one-component system phase diagrams, two-component system phase diagrams, and three-component system phase diagrams.

- 相图：由实验研究系统中平衡相与所处条件（T、P、X)的关系，把实验结果绘制成的几何图形。
- 相图按照组分数可分为单组分系统相图、二组分系统相图、三组分系统相图。

2. Phase Equilibrium of One-Component System (单组分系统相平衡)

2.2 Phase Diagram of Water (水的相图)

- Classified according to the mutual-solubility behavior between components, phase diagrams can also be divided into those for systems with completely-miscible liquids, completely-immiscible liquids, and partially-miscible liquids; as well as those for systems with completely-miscible solids, completely-immiscible solids, and partially-miscible solids.
- According to the relationship between system properties and composition, phase diagrams are further categorized into vapor-pressure-composition diagrams, boiling-point-composition diagrams, melting-point-composition diagrams, and so on.
- 按照组分间相互溶解的情况，又可分为液态完全互溶系统相图、完全不互溶系统相图、部分互溶系统相图。以及固态完全互溶系统相图、 完全不互溶系统相图、 部分互溶系统相图等。
- 按照其系统的性质与组成的关系又可分为蒸汽压-组成图，沸点-组成图，熔点-组成图等。

2. Phase Equilibrium of One-Component System (单组分系统相平衡)

2.2 Phase Diagram of Water (水的相图)

① Preliminary analysis:

- One-Component System of Water

K=1, Φ=1 (solid, liquid or gas). f = 1-1+2 = 2 (T and P), a bivariant system.

① 初步分析:

- 水的单组分系统

初步分析： K = 1，Φ = 1，S或l或g，f =1-1+2 = 2 (T和P)，双变量系统

2.2 Phase Diagram of Water (水的相图)

- Two-Component System of Water

Preliminary analysis: K=1, Φ=2. f = 1-2+2 = 1 (T or P), a univariant system.

- 水的双组分系统

初步分析： K = 1，Φ = 2，f =1-2+2 = 1 (T或P)，单变量系统

2. Phase Equilibrium of One-Component System (单组分系统相平衡)

2.2 Phase Diagram of Water (水的相图)

- Three-Component System of Water

Preliminary analysis: K=1, Φ=3. f = 1-3+2 = 0, invariant system.

- 水的三组分系统

初步分析： K = 1，Φ = 2，f =1-3+2 = 0，无变量系统

Preliminary analysis:

② Experimental Determination

Measure the specific relationship of P=f(T) under two-phase equilibrium conditions.

② 实验测定

在两相平衡时，测定 P= f (T) 的具体关系。

2. Phase Equilibrium of One-Component System (单组分系统相平衡)

2.2 Phase Diagram of Water (水的相图)

Phase-Equilibrium Data of Water

temperature / ℃	Vapor pressure of the system/KPa		Equilibrium external pressure
	Water to gas Gas to water	Ice to gas Gas to ice	Ice to water Water to ice
- 20	—	0.103	199.6×10^3
- 15	0.191	0.1651	161.1×10^3
- 10	0.285	0.2594	114.5×10^3
- 5	0.421	0.4010	61.8×10^3
0.0098	0.6106	0.6106	0.6106
20	2.338		
100	101.325		
374	22.038×10^3		

2. Phase Equilibrium of One-Component System (单组分系统相平衡)

2.2 Phase Diagram of Water (水的相图)

It can be seen from the tabulated data:

For water-vapor equilibrium, the vapor pressure increases with rising temperature.

For ice-vapor equilibrium, the vapor pressure increases with rising temperature.

For ice-water equilibrium, the melting point of ice decreases as the external pressure increases.

At 0.0098 ℃ and 610 Pa, ice, liquid water and vapor coexist in three-phase equilibrium.

从表列数据可看出:

水-汽平衡，蒸气压随温度升高而增大。冰-汽平衡，蒸气压随温度升高而增大。冰-水平衡，外压增大，冰的熔点降低。在0.0098℃ 和 610Pa下，冰、水、汽呈三相平衡。

2. Phase Equilibrium of One-Component System (单组分系统相平衡)

2.2 Phase Diagram of Water (水的相图)

③ Plotting: Take P as the ordinate and T as the abscissa. Plot the corresponding points of the above-mentioned data on the coordinate plane, and then connect the points for two-phase equilibrium separately to obtain three curves OA, OB and OC.

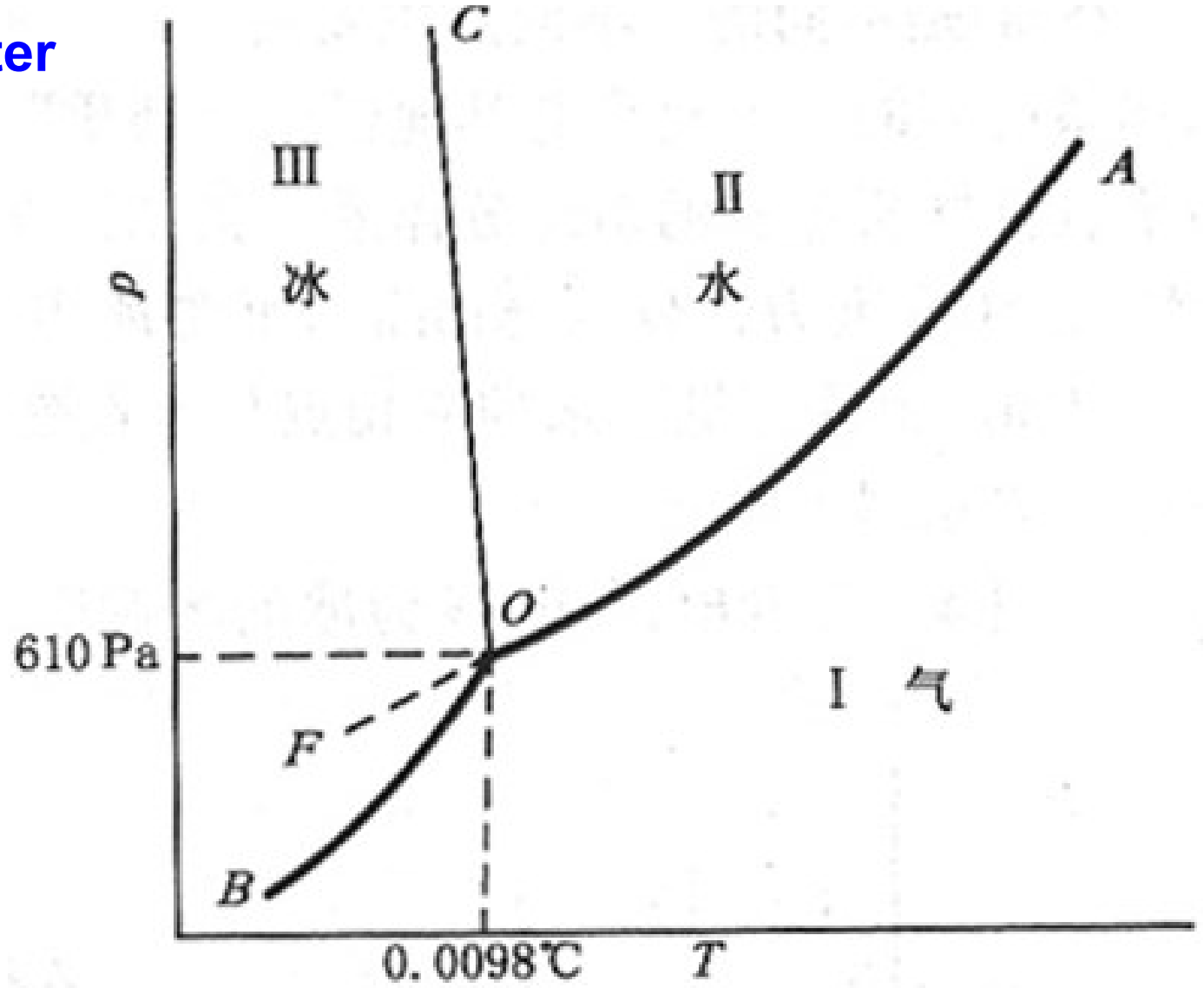
Point O is the intersection of the three curves. The three curves divide the coordinate plane into three regions I, II and III. Thus the phase diagram of water is obtained.

③ 作图：以P为纵座标，以T为横座标，将以上数据在座标平面上描出相应的点，再把两相平衡的点分别连成线，得 OA、OB、OC三曲线。三曲线的交点为O。三曲线将座标平面划分为I、II、III三个区域，便得到水的相图 。

2. Phase Equilibrium of One-Component System (单组分系统相平衡)

Phase Diagram of Water

水的相图



2.2 Phase Diagram of Water (水的相图)

④ Plotting (详细分析)

Every state of a system can be described by its temperature and pressure, and can be represented by a point on the phase diagram. Conversely, each point on the phase diagram corresponds to a certain state of the system.

系统的每一个状态都可以用系统所处的温度、压力来描述，可在相图上找到一个点来表示。反之，相图上每一个点也都对应着系统的某一状态。

2. Phase Equilibrium of One-Component System (单组分系统相平衡)

2.2 Phase Diagram of Water (水的相图)

④ Plotting (详细分析)

Curves: f =1-2+2=1

Curves OA, OB and OC represent liquid-gas, solid-gas, and liquid-solid two-phase equilibria respectively. These correspond to univariant systems for a one-component system with two phases in equilibrium, which are represented by lines on the phase diagram. Line OA: saturated vapor-pressure curve of water; Line OB: saturated vapor-pressure curve of ice; Line OC: melting curve of ice (or freezing curve of water).

曲线：f =1-2+2=1

曲线OA、OB、OC 分别表示液-气、固-气、液-固两相平衡，即单组分、两相平衡的单变量系统，在相图上表示为线。OA线：水的饱和蒸汽压曲线；OB线：冰的饱和蒸汽压曲线；OC线：冰的熔化线或水的凝固线。

2.2 Phase Diagram of Water (水的相图)

④ Plotting (详细分析)

Notes: 1) Each point on the curves represents a system in two-phase equilibrium; 2) With regard to the slopes of the three curves, the slope of line OB is greater than that of line OA, that also called $\left(\frac{dP}{dT}\right)_{升华} > \left(\frac{dP}{dT}\right)_{蒸发}$. This phenomenon can be explained by the Clausius-Clapeyron equation.

说明：1）线上每一点都代表一个两相平衡的系统；2）关于三条线的斜率，OB线的斜率 > OA线的斜率，即 $\left(\frac{dP}{dT}\right)_{升华} > \left(\frac{dP}{dT}\right)_{蒸发}$，这可由克-克方程解释。

2.2 Phase Diagram of Water (水的相图)

④ Plotting (详细分析)

$$\left(\frac{dP}{dT}\right)_{蒸发}=\frac{\Delta H_{m,蒸发}}{T\left(V_{气}-V_{液}\right)}\cong\frac{\Delta H_{m,蒸发}}{TV_{气}}$$

$$\left(\frac{dP}{dT}\right)_{升华}=\frac{\Delta H_{m,升华}}{T\left(V_{气}-V_{固}\right)}\cong\frac{\Delta H_{m,升华}}{TV_{气}}$$

$$\because \Delta H_{升华}=\Delta H_{熔化}+\Delta H_{蒸发} \qquad \therefore \Delta H_{升华}>\Delta H_{蒸发}$$

$$\therefore\left(\frac{dP}{dT}\right)_{升华}>\left(\frac{dP}{dT}\right)_{蒸发}$$

Since no additional assumptions are introduced in the deduction, the slope of the sublimation line is greater than the slope of the vaporization line for any one-component system.

由于推论中未引入任何条件，所以任何单组分系统的升华线的斜率都大于蒸发线的斜率。

2. Phase Equilibrium of One-Component System (单组分系统相平衡)

2.2 Phase Diagram of Water (水的相图)

⑤ Triple point and freezing point: The triple point corresponds to a strictly one-component system at 610 Pa and 0.0098 ℃. The freezing point refers to the two-phase equilibrium system of water and ice exposed to air. Dissolved impurities such as CO_2, O_2 and N_2 in water turn it into a multicomponent system, which lowers the freezing point by 0.00242 ℃ relative to the triple point.

⑤ 三相点和冰点：三相点是严格的单组分系统，即610Pa、0.0098 ℃；冰点是暴露在空气中的水、冰两相平衡系统。水中含有CO_2、O_2、N_2等杂质，成为多组分系统，使冰点较三相点下降了0.00242 ℃。

2. Phase Equilibrium of One-Component System (单组分系统相平衡)

2.2 Phase Diagram of Water (水的相图)

At an air pressure of 101325 Pa (including water vapor, CO_2, O_2, N_2 and other components), the freezing point is further lowered by 0.00747 ℃ relative to the triple point, giving a total depression of 0.0098 ℃. Therefore, the triple point of water is 0.0098 ℃, and the freezing point of water is 0 ℃.

空气为101325Pa（包括水蒸汽、CO_2、O_2、N_2等），又使冰点较三相点下降了0.00747℃，共下降了0.0098 ℃。所以：水的三相点是0.0098 ℃，水的冰点是0 ℃。

3. Completely-Miscible Binary-Liquid System (完全互溶的双液系统)

A binary-liquid system is defined as a system with the number of components K = 2, where both components exist in the liquid state. According to the phase rule, f = 2-Φ+2 = 4-Φ. The maximum degree of freedom f=3 (temperature, pressure, and composition) occurs when the number of phases reaches its minimum value of one. For convenient graphical representation, two-dimensional plots are employed. Under constrained conditions, f=2: a P-X diagram is obtained at constant temperature (T = constant), whereas a T-X diagram is obtained at constant pressure (P = constant).

组分数K = 2，并且两个组分皆为液态，称为双液系统。

f =2-Φ+2 = 4-Φ，相数最少为一相时，自由度最大：f = 3（温度、压力、组成）；为了便于书面表达，画平面图。f = 2：当T一定时为P-X图；当P-定时为T-X图。

3. Completely-Miscible Binary-Liquid System (完全互溶的双液系统)

3.1 P-X and T-X Diagrams of Ideal Solutions (理想溶液的P-X图及T-X图)

1) P-X diagram (vapor pressure-composition diagram)

① Liquid-phase line: For an ideal solution composed of components A and B at a given temperature, $P_i=P_i^*X_i$. The relationship between vapor pressure and composition of component A is as follows: $P_A=0$ when $X_A=0$; $P_A=P_A$ when $X_A=1$, which forms a straight line.

1）P-X图（蒸气压-组成图）

① 液相线：由A、B 两组分组成的理想溶液，在一定温度下，$P_i=P_i^*X_i$。组分A的蒸气压与组成的关系：当$X_A=0$ 时，$P_A=0$；当$X_A=1$ 时，$P_A=P_A^*$为一直线。

3. Completely-Miscible Binary-Liquid System (完全互溶的双液系统)

3.1 P-X and T-X Diagrams of Ideal Solutions (理想溶液的P-X图及T-X图)

1) P-X diagram (vapor pressure-composition diagram)

① Liquid-phase line: The relationship between vapor pressure and composition of component B: $P_B=0$ when $X_B=0$; PB=PB* when $X_B=1$, which is also a straight line. Total pressure: $P_{Total}= P_A+P_B=P_A^*X_A+P_B^*X_B= P_A^*(1-X_B)+P_B^*X_B=P_A^*-(P_A^*-P_B^*)X_B$.

1）P-X图（蒸气压-组成图）

① 液相线：组分B的蒸气压与组成的关系：当$X_B=0$时，$P_B=0$；当$X_B=1$时，$P_B=P_B^*$也为一直线。

$P_{总}= P_A+P_B=P_A^*X_A+P_B^*X_B= P_A^*(1-X_B)+P_B^*X_B=P_A^*-(P_A^*-P_B^*)X_B$。

3. Completely-Miscible Binary-Liquid System (完全互溶的双液系统)

3.1 P-X and T-X Diagrams of Ideal Solutions (理想溶液的P-X图及T-X图)

1) P-X diagram (vapor pressure-composition diagram)

① Liquid-phase line: When $X_B=0$, $P_{Total}=P_A^*$; when $X_B=1$, $P_{Total}=P_B^*$. Therefore, the relationship between P_{Total} and the solution composition is also a straight line. The total vapor pressure of an ideal solution lies between P_A^* and P_B^*.

Accordingly, the curve representing the relationship between the liquid-phase composition of the solution and the total vapor pressure is termed the liquid-phase line.

1）P-X图（蒸气压-组成图）

① 液相线：当$X_B=0$时，$P_{总}=P_A^*$，当$X_B=1$时，$P_{总}=P_B^*$。所以，$P_{总}$与溶液组成的关系也是一直线。理想溶液的$P_{总}$介于P_A^*~P_B^*之间。

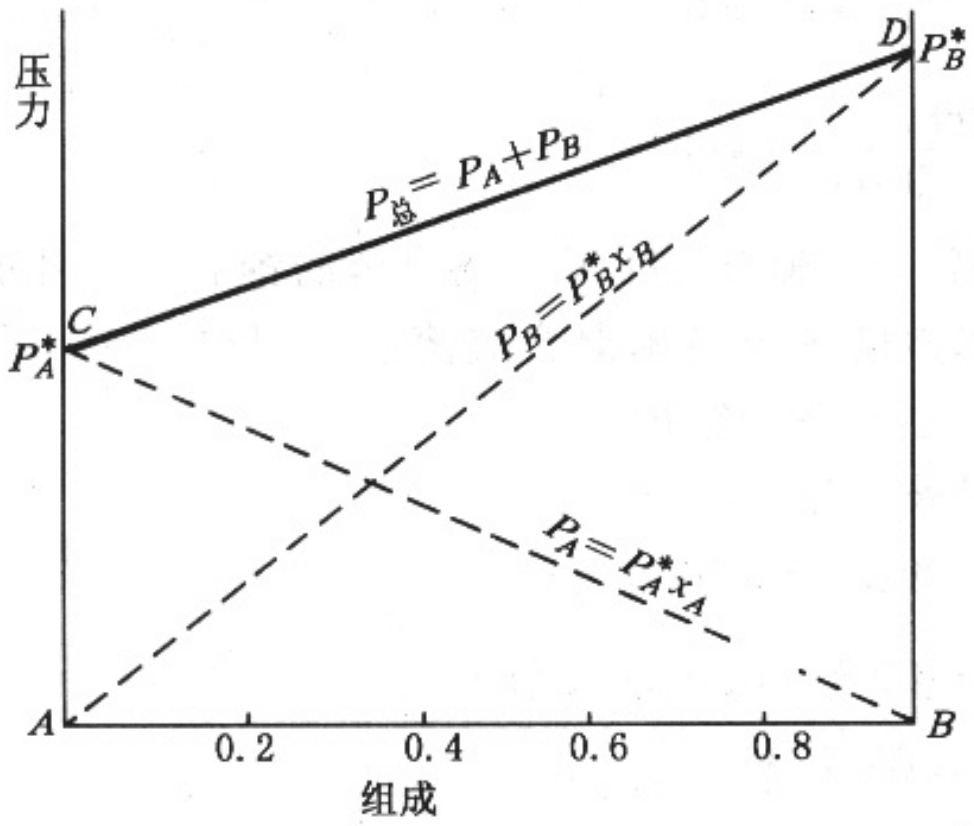
因此，把表示溶液的液相组成与蒸气总压之间关系的线称为液相线。

3. Completely-Miscible Binary-Liquid System (完全互溶的双液系统)

3.1 P-X and T-X Diagrams of Ideal Solutions (理想溶液的P-X图及T-X图)

1) P-X diagram (vapor pressure-composition diagram)

① Liquid-phase line:



P-X Diagrams for Ideal Solutions（理想溶液的P-X图）

3. Completely-Miscible Binary-Liquid System (完全互溶的双液系统)

3.1 P-X and T-X Diagrams of Ideal Solutions (理想溶液的P-X图及T-X图)

1) P-X diagram (vapor pressure-composition diagram)

② Vapor-phase line: Now consider the vapor in equilibrium with the solution. Let Y denote the mole fraction of components in the vapor phase. At relatively low vapor pressures, the vapor can be treated as an ideal gas.

$P_A = P_{Total} Y_A$； $P_A = P_A^* X_A$； $P_{Total} Y_A = P_A^* X_A$.

$$Y_A = \frac{P_A^* X_A}{P_{总}}$$

1）P-X图（蒸气压-组成图）

② 气相线：再讨论与溶液平衡的蒸气，以Y表示气相中物质的摩尔分数。蒸气压不高，可看成理想气体。$P_A = P_{总} Y_A$； $P_A = P_A^* X_A$； $P_{总} Y_A = P_A^* X_A$。

3. Completely-Miscible Binary-Liquid System (完全互溶的双液系统)

3.1 P-X and T-X Diagrams of Ideal Solutions (理想溶液的P-X图及T-X图)

1) P-X diagram (vapor pressure-composition diagram)

② Vapor-phase line: Similarly, at a given temperature, the corresponding vapor-phase composition Y_A or Y_B can be calculated from P_A, P_B, and the measured total pressure P_{Total} at different liquid-phase compositions X_A or X_B. The curve representing the relationship between total vapor pressure and vapor-phase composition is defined as the vapor-phase line.

1）P-X图（蒸气压-组成图）

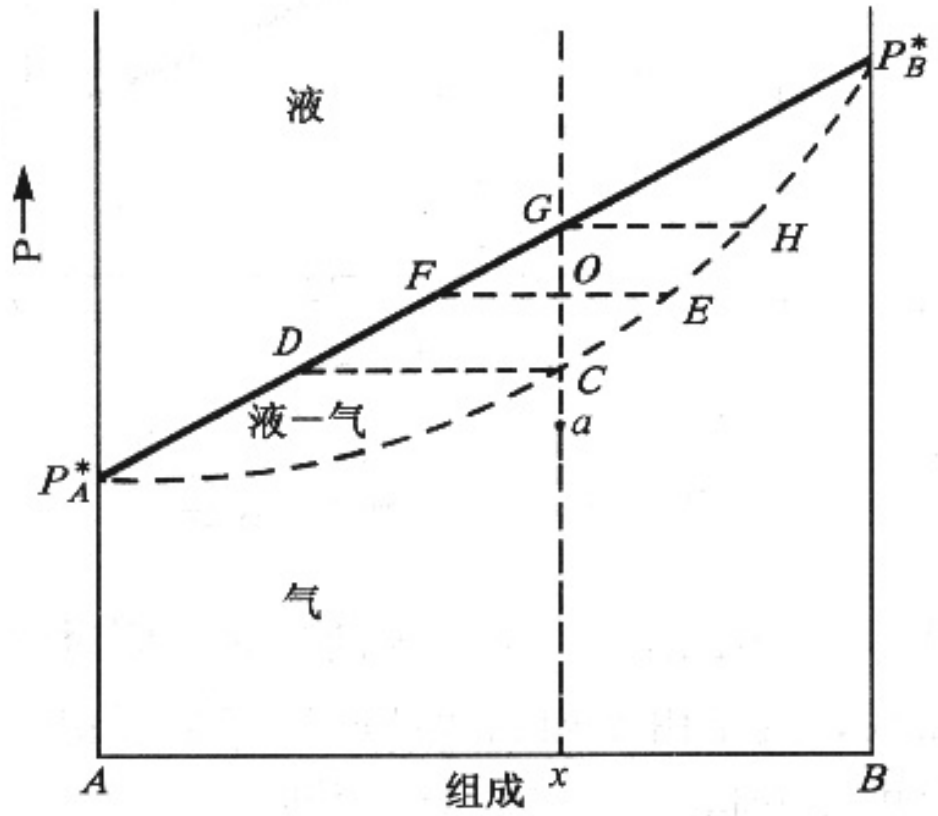
② 气相线：同理，$Y_B = \frac{P_B^* X_B}{P_{总}}$ ，在一定温度下，可由P_A^*、P_B^*及不同液相组成X_A（或X_B）时所测的$P_{总}$可计算出相应的气相组成Y_A（或Y_B），把表示蒸气总压与气相组成之间的关系线称为气相线。

3. Completely-Miscible Binary-Liquid System (完全互溶的双液系统)

3.1 P-X and T-X Diagrams of Ideal Solutions (理想溶液的P-X图及T-X图)

1) P-X diagram (vapor pressure-composition diagram)

② Vapor-phase line:



P-X Diagrams for Ideal Solutions（理想溶液的P-X图）

3. Completely-Miscible Binary-Liquid System (完全互溶的双液系统)

3.1 P-X and T-X Diagrams of Ideal Solutions (理想溶液的P-X图及T-X图)

1) P-X diagram (vapor pressure-composition diagram)

③ Relationship between liquid-phase composition and vapor-phase composition: If component B is more volatile than component A, namely $P_A<P_B$ then $P_A<P_{Total}<P_B$. Substituting into the above formula yields $Y_B>X_B$ or $Y_A<X_A$. That is, the more volatile component B has a higher content in the vapor phase than in the liquid phase. In the P-X diagram of an ideal solution, the vapor-phase line lies below the liquid-phase line.

$$Y_B = \frac{P_B^*}{P_{总}} X_B \qquad Y_A = \frac{P_A^*}{P_{总}} X_A$$

1）P-X图（蒸气压-组成图）

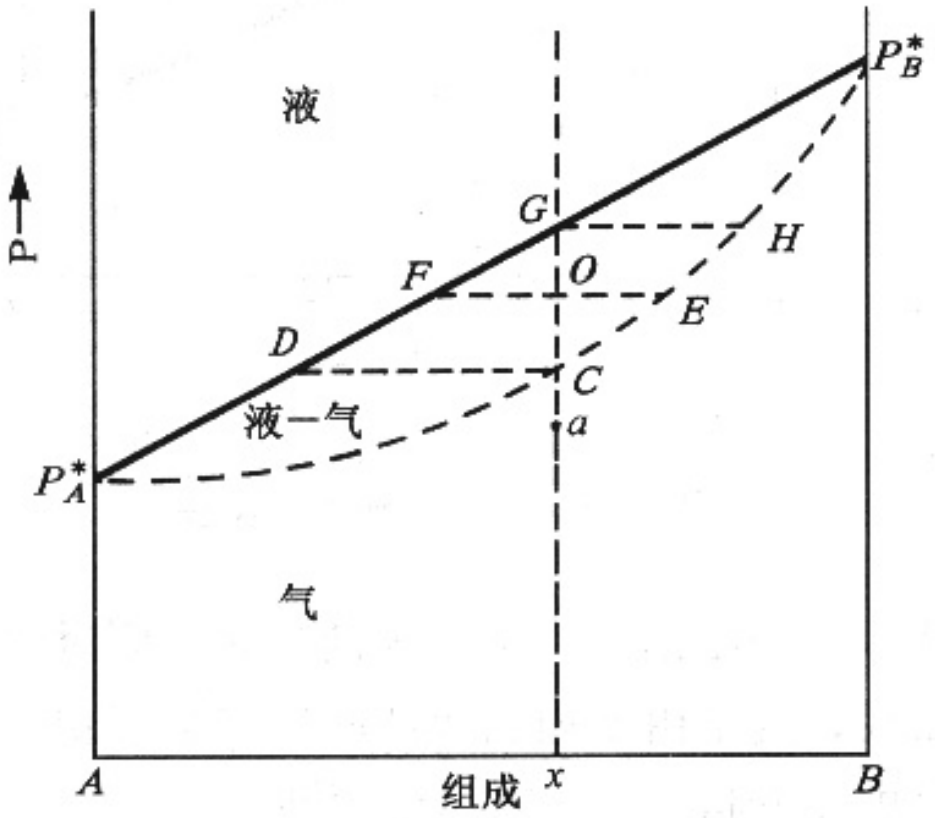
③液相组成与气相组成的关系：若B比A易挥发，即$P_A<P_B$，则$P_A<P_{总}<P_B$，带入上式得到$Y_B>X_B$；$Y_A<X_A$，即易挥发组分B在气相中的含量大于液相中的含量。理想溶液的P-X图中，气相线在液相线的下方。

3. Completely-Miscible Binary-Liquid System (完全互溶的双液系统)

3.1 P-X and T-X Diagrams of Ideal Solutions (理想溶液的P

1) P-X diagram (vapor pressure-composition diagram)

④ Applications of the P-X diagram: The solid straight line in which represents the relationship between liquid-phase com the relationship between X_B and P_{Total}. The dashed curve is the the relationship between vapor-phase composition and tota between Y_B and P_{total}.



1）P-X图（蒸气压-组成图）

④ P-X图的应用：图中实线（直线）为液相线，表示液相组成与$P_{总}$的关系，即X_B~$P_{总}$的关系。虚线（曲线）为气相线，表示气相组成与$P_{总}$的关系，即Y_B~$P_{总}$的关系。

3. Completely-Miscible Binary-Liquid System (完全互溶的双液系统)

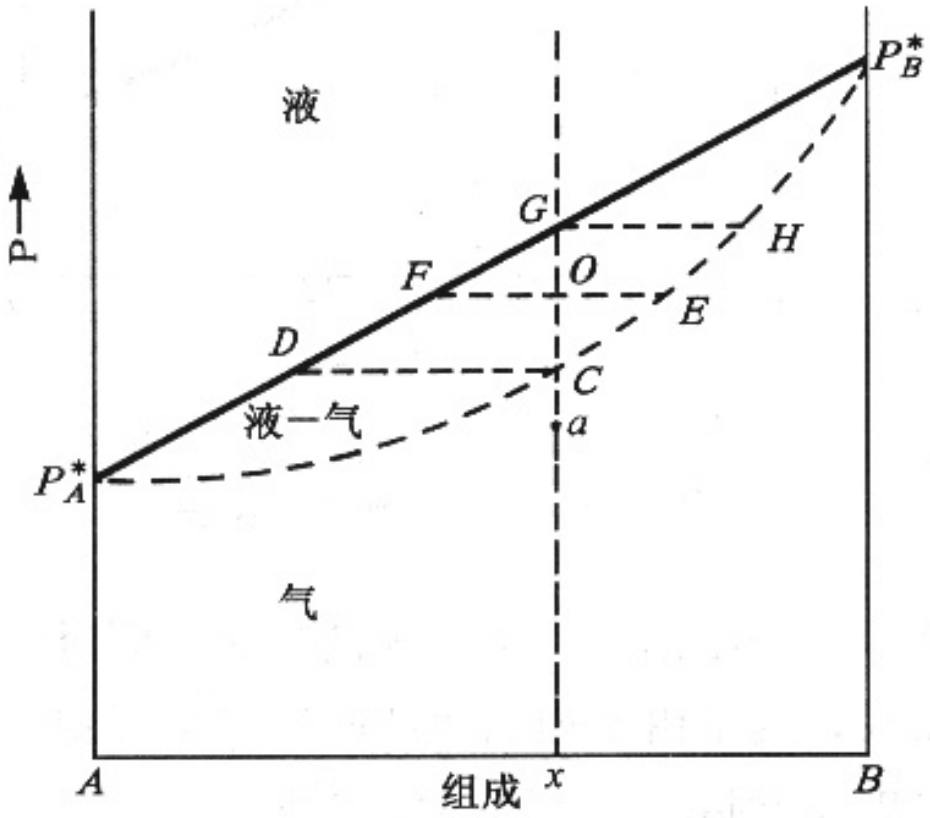
3.1 P-X and T-X Diagrams of Ideal Solutions (理想溶液的P

1) P-X diagram (vapor pressure-composition diagram)

④ Applications of the P-X diagram: Above the liquid-phase lin

f = 2-1+1 = 2. Below the vapor-phase line lies the vapor-phase

between the two lines is the gas-liquid two-phase coexistence



1）P-X图（蒸气压-组成图）

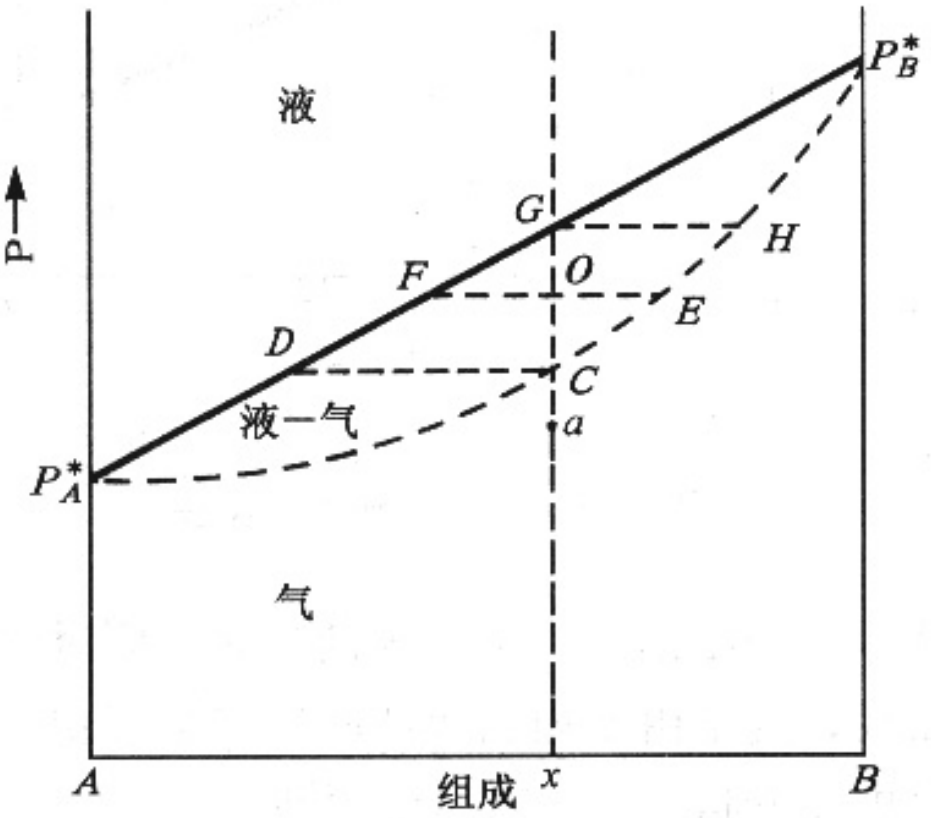
④ P-X图的应用： 液相线的上方为液相区，f =2-1+1=2；气相线的下方为气相区，f =2-1+1=2。两线中间为气、液两相共存区，f=2-2+1=1。

3. Completely-Miscible Binary-Liquid System (完全互溶的双液系统)

3.1 P-X and T-X Diagrams of Ideal Solutions (理想溶液的P

1) P-X diagram (vapor pressure-composition diagram)

④ Applications of the P-X diagram: The system point is the pc
the overall composition and pressure (or temperature) of the
phase points are the points in a phase diagram that denote
temperature) of each phase at equilibrium in the system (Point



1）P-X图（蒸气压-组成图）

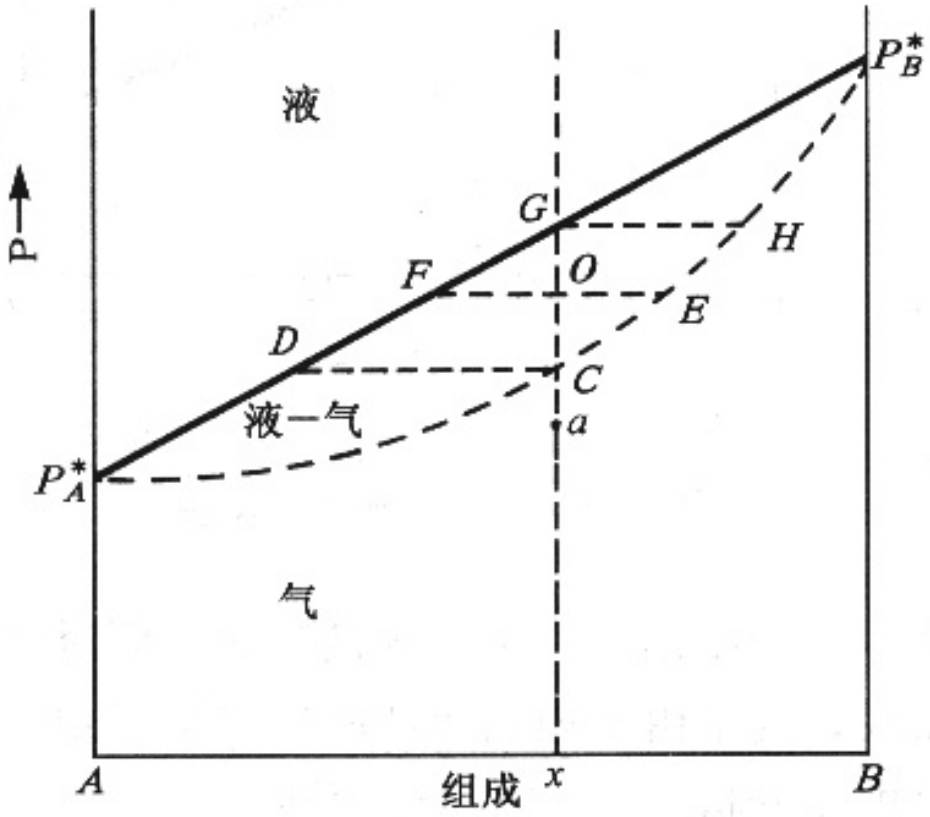
④ P-X图的应用：物系点是在相图中表示系统的总组成及压力（或温度）的点（图中O点）；相点是在相图中，表示系统平衡各相的组成及压力（或温度）的点（图中E、F点）。

3. Completely-Miscible Binary-Liquid System (完全互溶的双液系统)

3.1 P-X and T-X Diagrams of Ideal Solutions (理想溶液的P

1) P-X diagram (vapor pressure-composition diagram)

④ Applications of the P-X diagram: In the single-phase region, the point, meaning the overall composition of the system is identical to the two-phase equilibrium region, the system point does not coinc overall composition of the system differs from the compositions of th since the total pressure of the system is fixed, the system point an horizontal line. The tie-line is the line connecting the phase points o the figure).



1）P-X图（蒸气压-组成图）

④ P-X图的应用：在单相区，系统的物系点和相点重合，即系统的总组成与该相的组成是相同的。在两相平衡区，系统的物系点和相点不重合，即系统的总组成与平衡两相的组成是不相同的。但是由于系统的总压一定，所以，系统的物系点和两个相点在同一条水平线上。结线即连接平衡两相相点的线（图中的EF线）。

3. Completely-Miscible Binary-Liquid System (完全互溶的双液系统)

3.1 P-X and T-X Diagrams of Ideal Solutions (理想溶液的P-X图及T-X图)

2) T-X Diagram (Boiling-point-composition Diagram)

At a given pressure, the temperature at which the vapor pressure of a two-component solution equals the external pressure is defined as its boiling point. For two-component systems with different compositions, the total vapor pressure varies, and thus their boiling points are different. The phase diagram that illustrates the relationship between boiling point and composition for a two-component system is known as a boiling-point-composition diagram.

2）T-X图（沸点-组成图）

在一定压力下，两组分溶液的蒸气压与外压相等的温度称为沸点。当两组分系统的组成不同时，蒸气总压不同，所以沸点不相同。表示两组分系统的沸点与组成关系的相图叫沸点-组成图。

3. Completely-Miscible Binary-Liquid System (完全互溶的双液系统)

3.2 P-X and T-X Diagrams of Non-Ideal Solutions (非理想溶液的P-X图及T-X图)

1) Deviations:

① Positive deviation: The vapor partial pressures of each component in the solution are all greater than the values calculated by Raoult's law.

a) Slight positive deviation: $P_B < P_{Total} < P_A$

b) Large positive deviation: A maximum value appears in P_{Total}

1) 偏差

①正偏差：溶液中各组分的蒸气分压，均大于拉乌尔定律的计算值。

a）正偏差较小，$P_B < P_{总} < P_A$

b）正偏差较大，$P_{总}$出现最大值

3. Completely-Miscible Binary-Liquid System (完全互溶的双液系统)

3.2 P-X and T-X Diagrams of Non-Ideal Solutions (非理想溶液的P-X图及T-X图)

1) Deviations：

② Negative deviation: The vapor partial pressures of each component in the solution are all smaller than the values calculated by Raoult's law. a) Slight negative deviation: $P_B<P_{Total}<P_A$; b) Large negative deviation: A minimum value appears in P_{Total}.

③ Causes: a) Association and dissociation; b) Formation of associated molecules; c) Differences in intermolecular forces. In a two-component system, when component A exhibits positive deviation, component B also exhibits positive deviation.

1) 偏差

②负偏差：溶液中各组分的蒸气分压均小于拉乌尔定律的计算值

a）负偏差较小，$P_B< P_{总} < P_A$；b）负偏差较大，$P_{总}$出现最小值

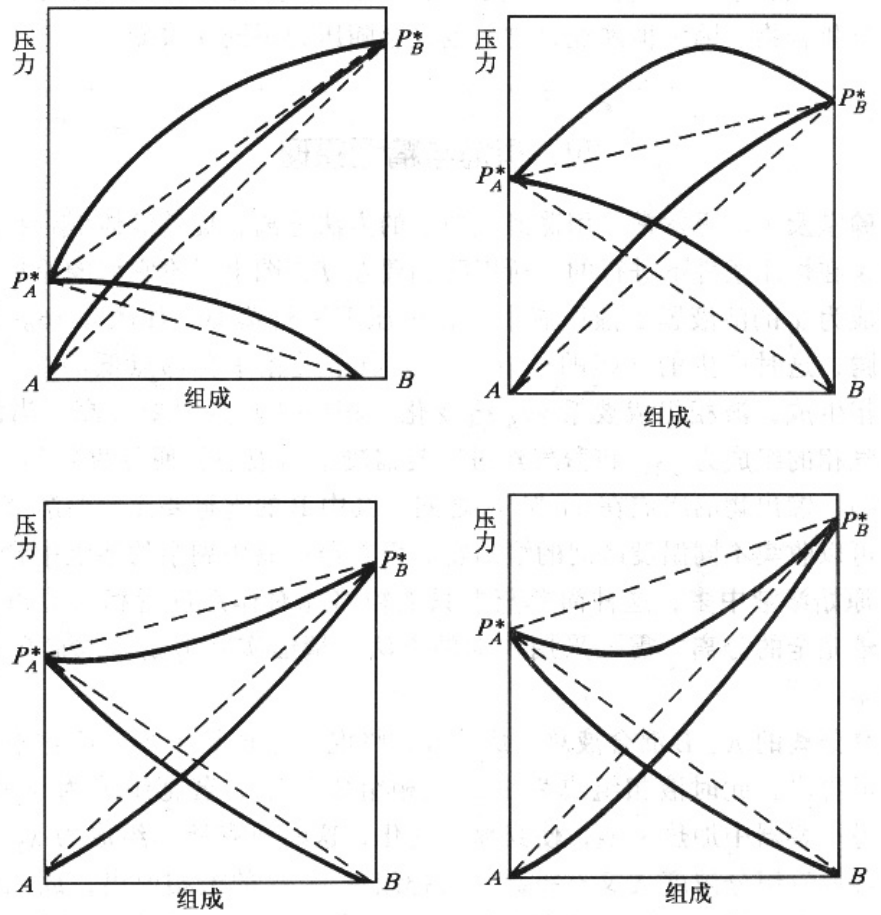
③原因： a) 缔合离解；b) 形成缔合分子；c) 分子间力不同

二组分系统中，组分A发生正偏差时，组分B也发生正偏差

3. Completely-Miscible Binary-Liquid System (完全互溶的双液系统)

3.2 P-X and T-X Diagrams of Non-Ideal Solutions (非理想溶液的P-X图及T-X图)

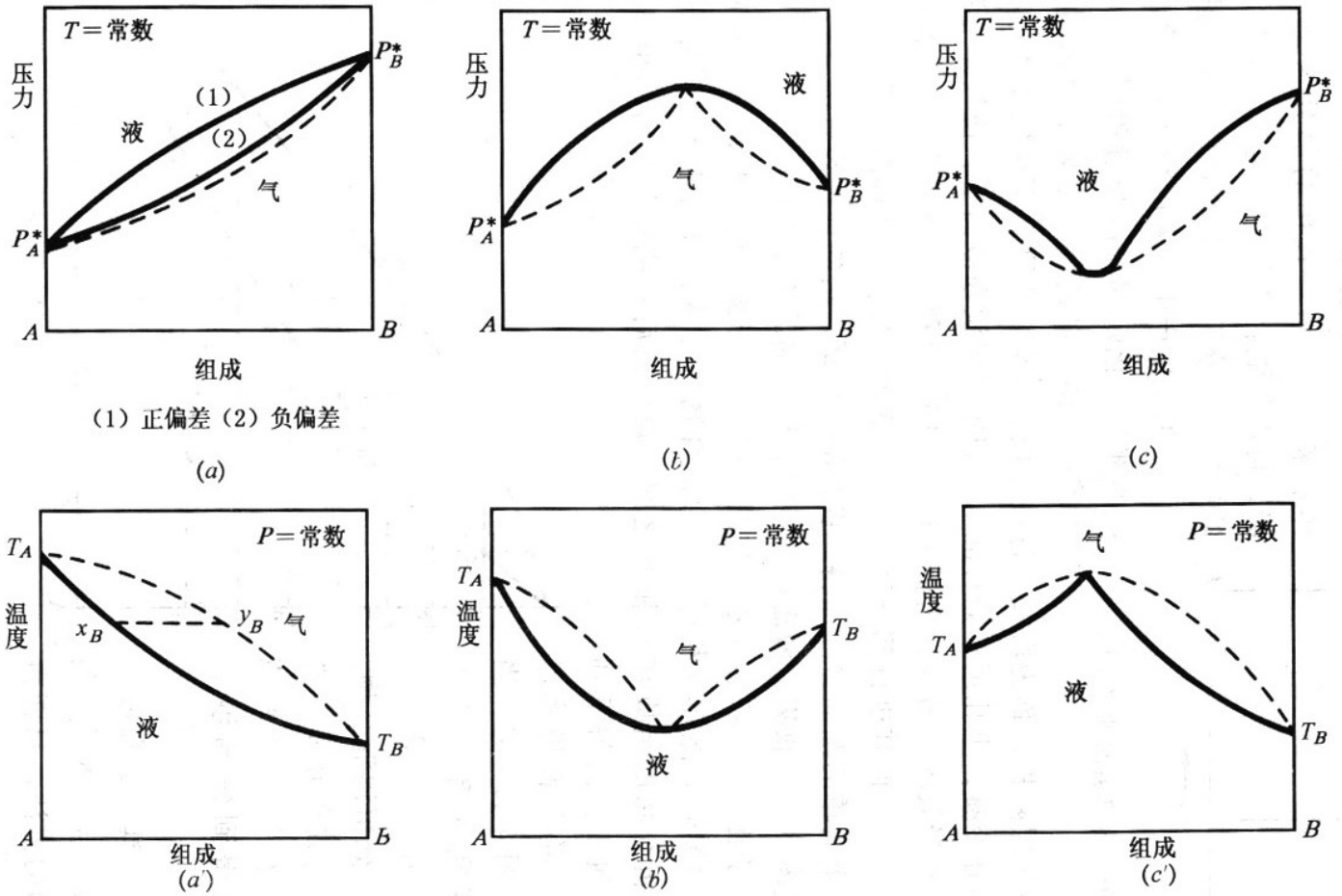
2) Phase Diagrams for Deviations of Non-Ideal Solutions （非理想溶液的偏差）



3. Completely-Miscible Binary-Liquid System (完全互溶的双液系统)

3.2 P-X and T-X Diagrams of Non-Ideal Solutions (非理想溶液的P-X图及T-X图)

2) Example for P-X and T-X Diagrams of Non-Ideal Solutions (非理想溶液的P-X及T-X图示例)



3. Completely-Miscible Binary-Liquid System (完全互溶的双液系统)

3.2 P-X and T-X Diagrams of Non-Ideal Solutions (非理想溶液的P-X图及T-X图)

2) Analysis for P-X and T-X Diagrams of Non-Ideal Solutions (非理想溶液的P-X及T-X图分析)

a) The T-X phase diagrams for ordinary positive and negative deviations are basically similar to those of ideal solutions; b) For phase diagrams with maximum positive or negative deviations, at the maximum (or minimum) point on the P-X diagram, the vapor-phase curve intersects the liquid-phase curve, indicating that the vapor-phase composition is identical to the liquid-phase composition of the system at this point; c) Maximum (minimum) azeotropic point: the maximum (minimum) point appearing on the T-X diagram; d) The composition corresponding to this point is referred to as the composition of the azeotropic mixture; e) The composition at the maximum (minimum) point on the P-X diagram is not necessarily the same as that at the minimum (maximum) point on the T-X diagram. The temperature and composition of the azeotropic point vary with external pressure.

a）具有一般正、负偏差T-X相图与理想溶液的T-X相图基本相似；b）具有最大正、负偏差时的相图，在P-X图上最高（或低）点处，气相线与液相线相交，说明此时系统的气相组成与液相组成相同；c）最高（低）恒沸点：在T-X图上出现的最高（低）点；d）该点对应的组成，称为恒沸混合物的组成；e）在P-X图上最高（低）的组成不一定与T-X图上最低（高）点的组成一样。恒沸点的温度及组成会随外压而变化。

3. Completely-Miscible Binary-Liquid System (完全互溶的双液系统)

3.3 Lever Rule (杠杆规则)

Definition: When the system point falls within the two-phase equilibrium region, the system exists in two-phase equilibrium. The quantity ratio of the two equilibrium phases can be determined from the positions of the system point and the two phase points on the phase diagram.

定义：当系统的物系点落在两相平衡区时，则系统呈两相平衡。可以根据物系点及两个相点在相图上的位置，确定平衡两相的数量比。

3. Completely-Miscible Binary-Liquid System (完全互溶的双液系统)

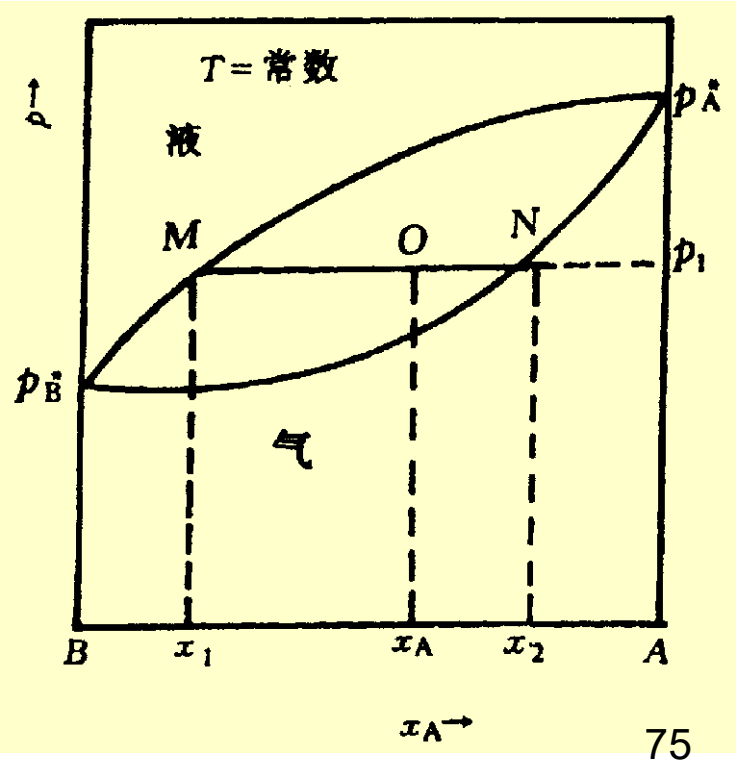
3.3 Lever Rule (杠杆规则)

Material balance for component A: $n_{Total}x_A = n_e x_1 + n_g x_2$； $(n_e+n_g)x_A = n_e x_1 + n_g x_2$,

Rearranging gives

$$n_e \cdot \overline{oM} = n_g \cdot \overline{oN}$$

$$\frac{n_e}{n_g} = \frac{\overline{oN}}{\overline{oM}}$$



对A进行物料衡算：$n_{总}x_A = n_e x_1 + n_g x_2$； $(n_e+n_g)x_A = n_e x_1 + n_g x_2$，

整理$n_e(x_A-x_1) = n_g(x_2-x_A)$，即

$$n_e \cdot \overline{oM} = n_g \cdot \overline{oN}$$

$$\frac{n_e}{n_g} = \frac{\overline{oN}}{\overline{oM}}$$

3. Completely-Miscible Binary-Liquid System (完全互溶的双液系统)

3.4 Principle of Rectification (精馏原理)

1) By means of rectification, substances of relatively high purity can be separated from the mixture of A and B, and A and B can even be completely separated into pure components. This is because the vapor in equilibrium with the binary-component solution of A and B contains a higher fraction of the volatile component than the liquid phase.

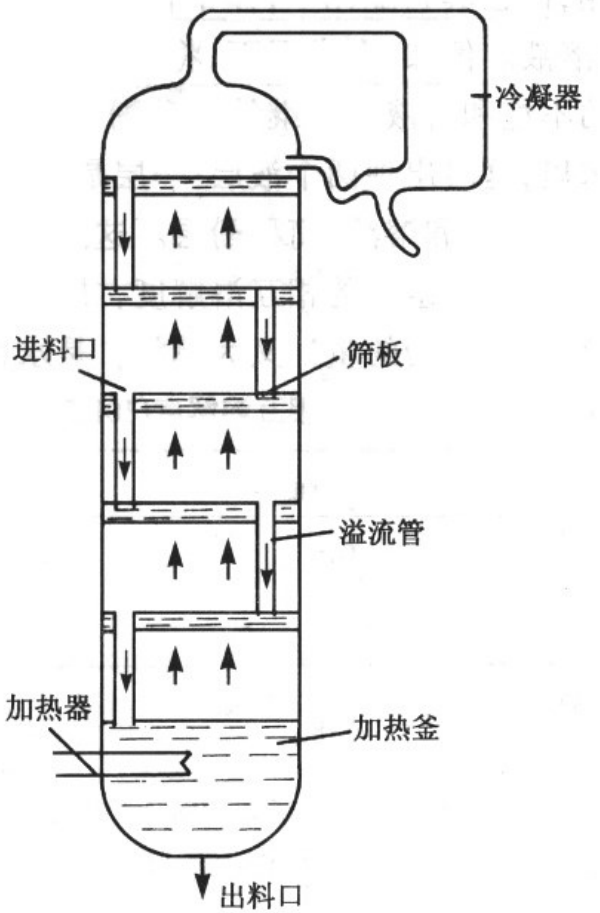
2) For a completely miscible binary-liquid system, continuous partial condensation of the vapor or continuous partial vaporization of the liquid phase can enrich the volatile component in the vapor phase and enrich the non-volatile component in the liquid phase. Through a series of such partial vaporization and partial-condensation operations, the mixture of A and B can be completely separated.

1) 利用精馏的方法，可以从A、B混合物中分离出纯度较高的物质，甚至可以将A、B完全分离为纯组分。这是由于与A、B两组分溶液相平衡的蒸气中，易挥发组分的含量比液相中多。

2) 对于完全互溶的双液系统，把气相不断的部分冷凝，或将液相不断的部分汽化都能在气相中浓集易挥发组分，在液相中浓集难挥发组分。这样进行一连串的部分汽化与部分冷凝，可将混合物A、B进行完全分离。

3. Completely-Miscible Binary-Liquid System (完全互溶的双液系统)

3.4 Principle of Rectification (精馏原理)



In industry and laboratories, rectification columns are commonly employed to realize such continuous-process operations.

工业上和实验室中，通常采用精馏塔和精馏柱实现这类连续过程。

Schematic Diagram of Rectification Column (精馏塔示意图)

4. Partially Miscible and Completely Immiscible Binary-Liquid Systems (部分互溶及完全不互溶的双液系统)

4.1 Phase Diagram of Partially Miscible Binary-Liquid System-Solubility Diagram (部分互溶的双液系统相图-溶解度图)

When two liquids differ significantly in their properties, partial miscibility may occur. Within a certain temperature range, the mutual solubility of the two liquids is low. A homogeneous single phase unsaturated solution can only form when one liquid is present in a very small amount. At other composition ratios, the system separates into two layers, with two saturated liquid solutions coexisting in phase equilibrium.

当两种液体的性质差别较大时，可发生部分互溶现象，即在某温度范围内两种液体相互溶解度都不大，只有当一种液体的量很少时才能形成均匀的一相，为不饱和溶液。在其它配比下，系统将分层而呈两个饱和溶液的液相平衡共存。

4. Partially Miscible and Completely Immiscible Binary-Liquid Systems (部分互溶及完全不互溶的双液系统)

4.1 Phase Diagram of Partially Miscible Binary-Liquid System-Solubility Diagram (部分互溶的双液系统相图-溶解度图)

For example: At 23.9 °C, when phenol is added to water, it dissolves completely at low concentrations and reaches saturation at 7.5 %. Further addition of phenol causes the system to separate into two layers: a saturated solution of water in phenol and a saturated solution of phenol in water. These two liquid phases are termed conjugate solutions. The concentrations of the two liquid layers are measured separately, which represent the mutual solubilities of the two components at this temperature. By determining the compositions of conjugate solutions at different temperatures, a temperature-composition diagram, also known as a solubility diagram, can be plotted.

例如：在温度 23.9℃时，往水中加苯酚，少量时完全溶解，至7.5%达饱和。继续加苯酚，分为两层：水在苯酚中的饱和溶液 ；苯酚在水中的饱和溶液，这两液相称为共轭溶液。分别测定两液层的浓度，即在该温度下，两组分的相互溶解度。测定不同温度下共轭溶液的组成，便可绘出其温度—组成图，又称为溶解度图。

4. Partially Miscible and Completely Immiscible Binary-Liquid Systems (部分互溶及完全不互溶的双液系统)

4.1 Phase Diagram of Partially Miscible Binary-Liquid System-Solubility Diagram (部分互溶的双液系统相图-溶解度图)

Mutual Solubility of Phenol and Water (苯酚和水的相互溶解度)

Temperature / ℃	Aqueous layer / mass % of phenol	Phenol-rich layer / mass % of phenol
2.6	6.9	75.6
23.9	7.5	71.2
29.6	7.8	70.7
32.5	8.0	69.0
38.8	8.8	66.6
45.7	9.7	64.6
50.0	11.5	62.0
55.5	12.0	60.0
59.8	13.6	57.7
60.5	14.0	55.5
61.8	15.0	54.0
65.0	18.5	50.0

4. Partially Miscible and Completely Immiscible Binary-Liquid Systems (部分互溶及完全不互溶的双液系统)

4.1 Phase Diagram of Partially Miscible Binary-Liquid System-Solubility Diagram (部分互溶的双液系统相图-溶解度图)

① Line ac represents the solubility curve of B in A; line bc represents the solubility curve of A in B.

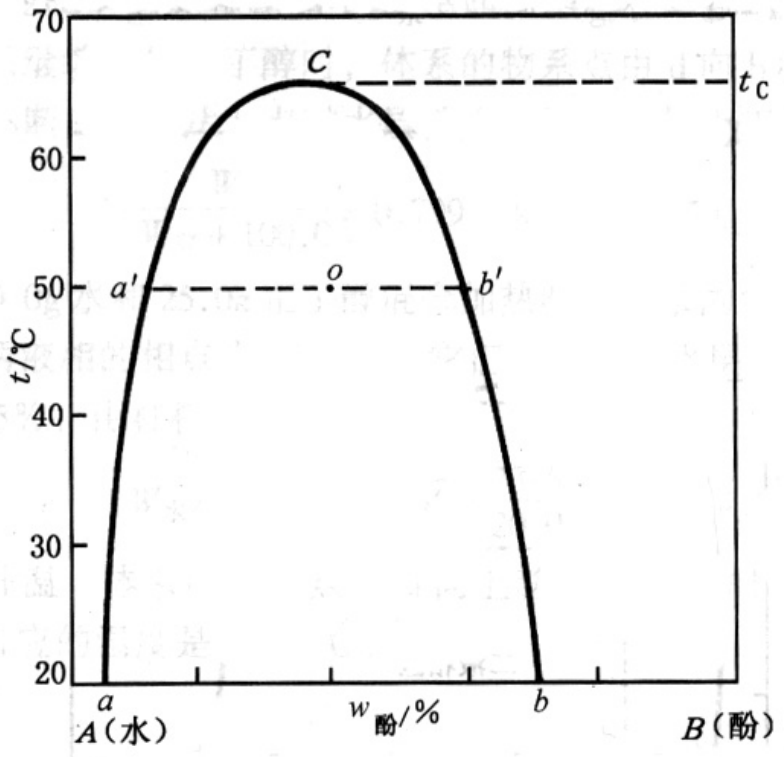
② The region outside curve acb is a single-liquid-phase region; the region inside curve acb is a two-liquid-phase coexistence region.

③ Point c is the upper critical solution temperature.

① ac线为B在A中的溶解度曲线；bc线为A在B中的溶解度曲线

② 曲线acb以外面为单一液相区；曲线acb以内面为两液相共存区

③ c点为最高临界溶解温度



Solubility Diagram of Partially Miscible Systems
部分互溶系统的溶解度图

4. Partially Miscible and Completely Immiscible Binary-Liquid Systems (部分互溶及完全不互溶的双液系统)

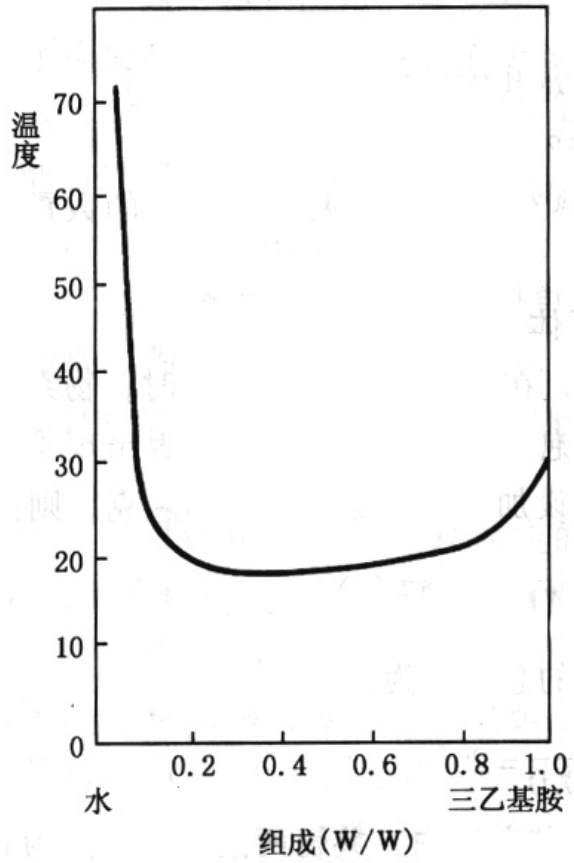
4.1 Phase Diagram of Partially Miscible Binary-Liquid System-Solubility Diagram (部分互溶的双液系统相图-溶解度图)

For some partially miscible systems, mutual solubility decreases as temperature rises, exhibiting a lower critical solution temperature. Certain partially miscible systems possess both an upper critical solution temperature and a lower critical solution temperature.

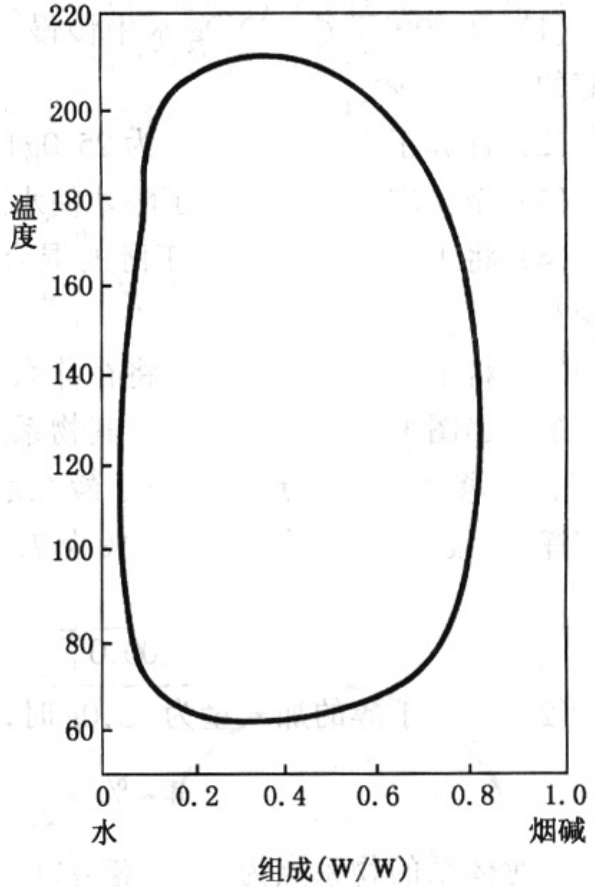
有的部分互溶系统当温度升高时，相互溶解度反而降低，有最低临界溶解温度。还有的部分互溶系统既有最高临界溶解温度又有最低临界溶解温度。

4. Partially Miscible and Completely Immiscible Binary-Liquid Systems (部分互溶及完全不互溶的双液系统)

4.1 Phase Diagram of Partially Miscible Binary-Liquid System-Solubility Diagram (部分互溶的双液系统相图-溶解度图)



Solubility Diagram with Lower Critical Solution Temperature
具有最低临界溶解温度的溶解度图



Solubility Diagram with Both Upper and Lower Critical Solution Temperatures
同时具有最高、低临界溶解温度的溶解度图

4. Partially Miscible and Completely Immiscible Binary-Liquid Systems (部分互溶及完全不互溶的双液系统)

4.2 Completely Immiscible Binary-Liquid System (完全不互溶的双液系统)

1) Phase diagram: When two liquids differ greatly in properties, their mutual solubility is so small that it can be neglected, and the system is regarded as completely immiscible, such as mercury-water and alkane-water systems. In such systems, each component exerts no influence on the other. At phase equilibrium, the vapor pressure of each component equals its saturated vapor pressure when it exists alone.

1) 相图：当两种液体性质差别非常大时，它们的相互溶解度很小，以致可以忽略不计，看成是完全不互溶的系统，例如汞和水；烷烃和水等。在这种系统中，各组分互不影响，达相平衡时，它们的蒸气压就是它们单独存在时的饱和蒸气压。

4. Partially Miscible and Completely Immiscible Binary-Liquid Systems (部分互溶及完全不互溶的双液系统)

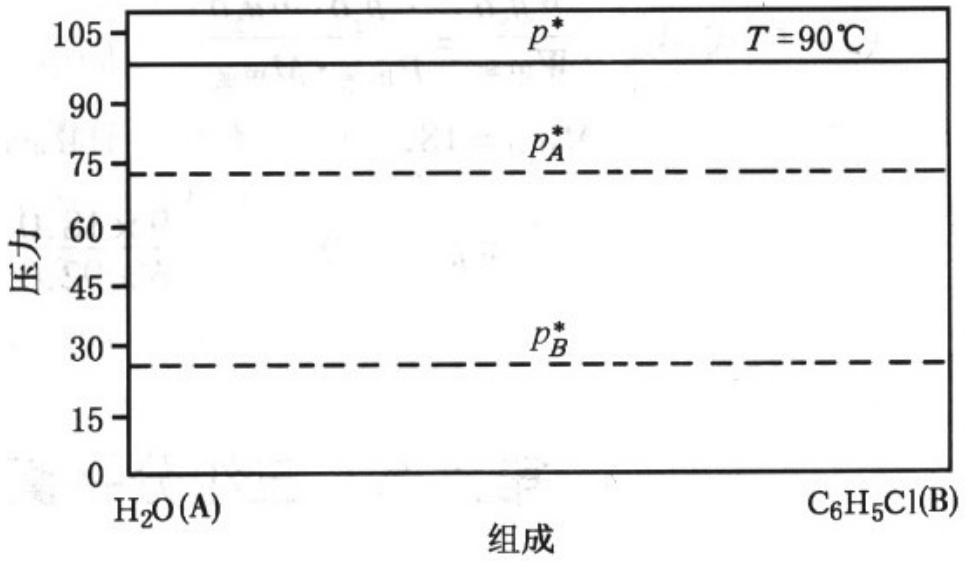
4.2 Completely Immiscible Binary-Liquid System (完全不互溶的双液系统)

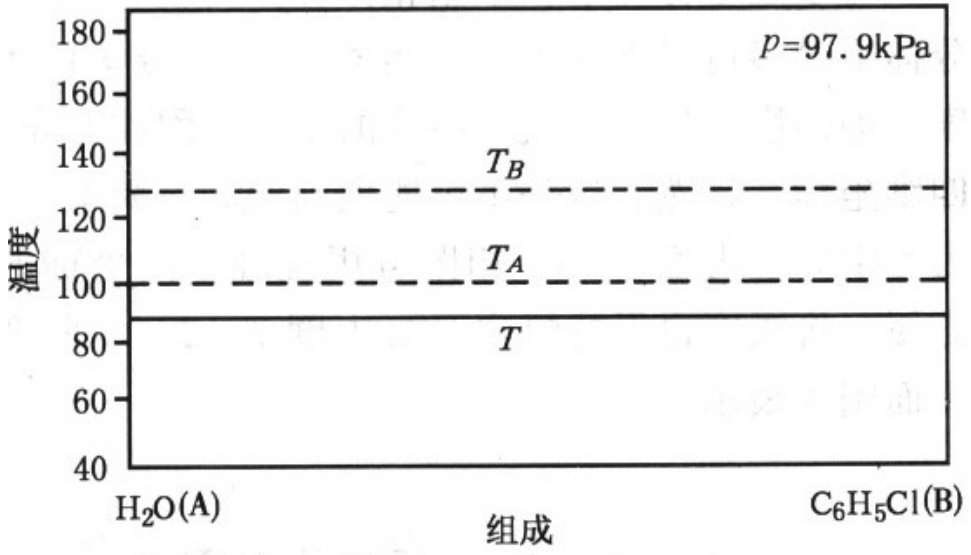
- It is only a function of temperature and independent of the amount of the other liquid component. Thus, $P_{Total} = P_A + P_B$. Since the total vapor pressure of the system is higher than the vapor pressure of each component at the same temperature, the boiling point of the system is lower than the boiling point of either pure component.

- 它只是温度的函数与另一液体组分的数量多少无关，则$P_{总} = P_A + P_B$，由于系统的蒸气总压 均高于同温度下各组分的蒸气压，所以系统的沸点 低于任一纯组分的沸点。

4. Partially Miscible and Completely Immiscible Binary-Liquid Systems (部分互溶及完全不互溶的双液系统)

4.2 Completely Immiscible Binary-Liquid System (完全不互溶的双液系统)





P-X and T-X Diagrams for Binary Immiscible Systems

二组分不互溶系统的P-X图及T-X图

4.2 Completely Immiscible Binary-Liquid System (完全不互溶的双液系统)

2) Steam Distillation: Taking advantage of the property that the boiling point of a completely immiscible binary-liquid system is lower than the boiling points of its pure components, water-insoluble organic compounds are often distilled together with water to avoid the decomposition of organic substances. This method is known as steam distillation.

2) 水蒸气蒸馏：利用完全不互溶双液系统的沸点低于各纯组分沸点的特性，常常将不溶于水的有机化合物和水一起蒸馏，避免有机物分解，这种方法称为水蒸气蒸馏。

5. Phase Diagrams of Binary Solid-Liquid Equilibrium Systems
(二组分固-液平衡系统的相图)

Solid-liquid equilibrium systems are condensed systems and are barely affected by pressure, so the pressure effect can be neglected, so f=2-Φ+1=3-Φ. At the minimum degree of freedom f=0, Φ=3, meaning a maximum of three phases can coexist. At the minimum number of phases Φ=1, f=2; accordingly, phase diagrams can be plotted using temperature-composition planar coordinates.

固-液平衡系统是凝聚系统，受压力影响很少，可不考虑，故f=2-Φ+1=3-Φ。当自由度最少时，f=0，则Φ=3，最多有3相。当相数最少时，Φ=1，则f=2，可用温度-组成平面坐标绘制相图。

5. Phase Diagrams of Binary Solid-Liquid Equilibrium Systems
(二组分固-液平衡系统的相图)

There are two methods for plotting phase diagrams of binary solid-liquid equilibrium systems:

① Thermal analysis: applied to alloy systems with relatively high melting points

② Solubility method: applied to salt-water systems where one phase is solid at room temperature

二组分固-液平衡系统相图的绘制方法有两种：

① 热分析法：应用于熔点较高的合金系统

②溶解度法：应用于常温下有一相为固相的盐水系统

5. Phase Diagrams of Binary Solid-Liquid Equilibrium Systems (二组分固-液平衡系统的相图)

5.1 Simple Eutectic Phase Diagram (简单低共熔相图；固态完全不互溶/液态完全互溶)

1) Thermal Analysis and Phase Diagrams

① Principle: Phase transitions are identified according to the temperature-time relationship during the cooling (or heating) process of the system.

②Method: Melt the sample completely and allow it to cool under a certain ambient condition (room temperature). Measure the data of temperature variation with time and plot the temperature-time curve, which is known as a cooling curve. Phase-transition temperatures can be identified from the turning points and plateaus on the curve. Plot cooling curves for samples with different compositions of the two components, obtain the phase-transition temperatures corresponding to various compositions, and the phase diagram can thereby be constructed.

1）热分析法及相图

① 原理：根据系统在冷却（或加热）过程中，温度随时间的变化关系，判断有无相变的发生。

② 方法：将样品全部熔化，让它在一定环境（室温）下冷却，测定温度随时间的变化数据，绘出温度 - 时间曲线叫步冷曲线，由曲线上转折、停顿点找出相变温度。将两组分不同组成的样品的步冷曲线绘出，找出不同组成时的相变温度，便可绘制相图。

5. Phase Diagrams of Binary Solid-Liquid Equilibrium Systems (二组分固-液平衡系统的相图)

5.1 Simple Eutectic Phase Diagram (简单低共熔相图；固态完全不互溶/液态完全互溶)

2) Solubility Method and Phase Diagrams

① Principle: Cooling an aqueous salt solution leads to the precipitation of solids. For dilute solutions, ice precipitates. The temperature at which ice separates out varies for dilute solutions of different concentrations, yet all are lower than the freezing point of pure water. For relatively concentrated solutions, solid salt precipitates. The precipitation temperature of salt differs for solutions of different concentrations, and the concentration (solubility) of the saturated solution coexisting with solid-state salt also varies.

② Method: A phase diagram can be constructed by measuring the freezing points of a series of salt-water solutions with different concentrations as well as the salt solubilities at various temperatures.

2）溶解度法及相图

① 原理：降低盐水溶液的温度，会有固体析出，当溶液较稀时，析出的是冰，且不同浓度稀溶液析出冰的温度不同，却都低于水的冰点。当溶液浓度较浓时，析出的是固体盐，不同浓度溶液析出盐的温度不同，且与固态盐共存的饱和溶液的浓度（溶解度）不同。

② 方法：测定一系列不同浓度盐水溶液的冰点及不同温度下盐的溶解度，便可绘制相图。

5. Phase Diagrams of Binary Solid-Liquid Equilibrium Systems (二组分固-液平衡系统的相图)

5.1 Simple Eutectic Phase Diagram (简单低共熔相图；固态完全不互溶/液态完全互溶)

3) The $(NH_4)_2SO_4$-H_2O system is taken as a representative example to illustrate the construction of solid-liquid phase diagrams by the solubility method (以$(NH_4)_2SO_4$-H_2O系统为代表展示溶解度法绘制固液相图)

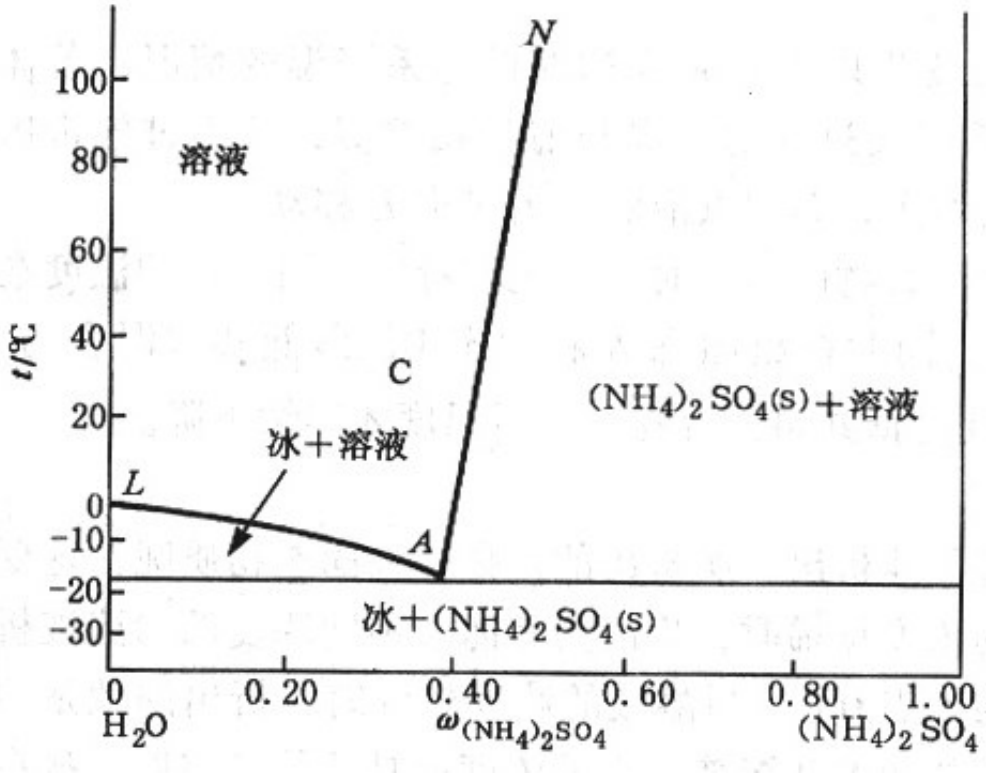
不同温度下$(NH_4)_2SO_4$-H_2O系统的相平衡数据

温度/℃	$(NH_4)_2SO_4$/%(W/W)	固相
- 5.55	16.7	冰
- 11	28.6	冰
- 18	37.5	冰
- 19.1	38.4	冰 + $(NH_4)_2SO_4$(s)
0	41.4	$(NH_4)_2SO_4$(s)
10	42.2	$(NH_4)_2SO_4$(s)
20	43.0	$(NH_4)_2SO_4$(s)
30	43.8	$(NH_4)_2SO_4$(s)
40	44.8	$(NH_4)_2SO_4$(s)
50	45.8	$(NH_4)_2SO_4$(s)
60	46.8	$(NH_4)_2SO_4$(s)

5. Phase Diagrams of Binary Solid-Liquid Equilibrium Systems (二组分固-液平衡系统的相图)

5.1 Simple Eutectic Phase Diagram (简单低共熔相图；固态完全不互溶/液态完全互溶)

3) The $(NH_4)_2SO_4$-H_2O system is taken as a representative example to illustrate the construction of solid-liquid phase diagrams by the solubility method (以$(NH_4)_2SO_4$-H_2O系统为代表展示溶解度法绘制固液相图)



① Line AN is the solubility curve of solid ammonium sulfate; line LA is the freezing-point depression curve of water;

② Point A represents the three-phase coexistence of ice, solid ammonium sulfate and solution. The corresponding temperature is the eutectic temperature, and the solid precipitated at this temperature is the eutectic mixture.

③ The line passing through point A and parallel to the horizontal axis is the three-phase line.

① 线AN为硫酸铵(s)的溶解度曲线；线LA为水的冰点下降曲线

② A点为冰、硫酸铵(s)和溶液三相共存，且所对应的温度为最低共熔点，在这一温度析出的固体是最低共熔混合物。

③ A点与横轴平行的线是三相线。

5.2 Examples of Applications of Simple Eutectic-System Phase Diagrams (简单低共熔系统相图应用举例)

1) Melting-point determination for sample purity inspection: For a synthetic drug, measure its melting point and compare it with that of a standard reference. If the melting points are identical, mix the sample with the standard reference and re-measure the melting point to confirm whether they are the same compound.

1) 测定熔点，检查样品的纯度：合成药品，测其熔点，与标准品比较，熔点相同，为了确定是同一种化合物，可以把样品和标准品混合后再测熔点。

5.2 Examples of Applications of Simple Eutectic-System Phase Diagrams (简单低共熔系统相图应用举例)

2) Laboratory Refrigeration (实验室致冷)

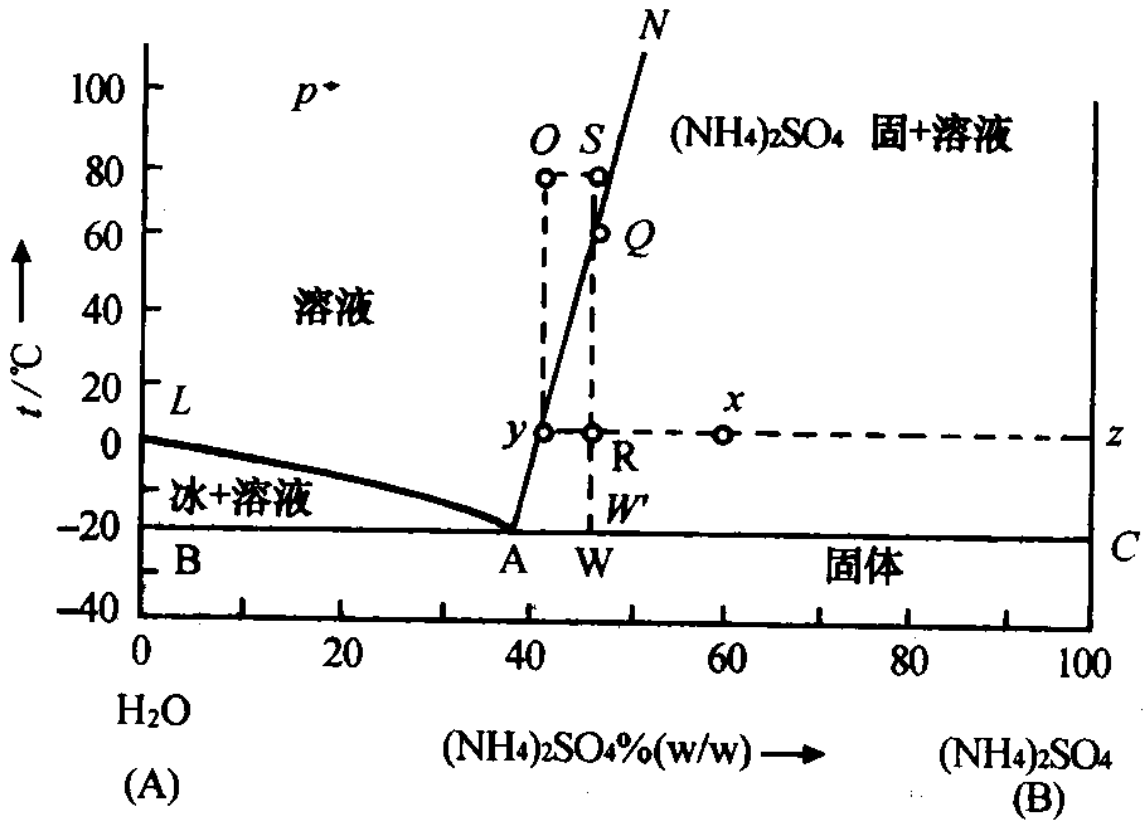
一些水-盐的低共熔混合物组成和最低共熔点
Compositions and eutectic temperatures of several water-salt eutectic mixtures

Salt（盐）	Eutectic temperature（最低共熔点）	Composition of eutectic mixture（低共熔混合物组成）
NaCl	-21.1 ℃	23.3 %
KCl	-10.7 ℃	19.7 %
NH_4Cl	-15.4 ℃	18.9 %
$CaCl_2$	-55 ℃	29.9 %
$(NH_4)_2SO_4$	-19.1 ℃	39.8 %

5. Phase Diagrams of Binary Solid-Liquid Equilibrium Systems (二组分固-液平衡系统的相图)

5.2 Examples of Applications of Simple Eutectic-System Phase Diagrams (简单低共熔系统相图应用举例)

3) Separation of solid mixtures and purification of salts by cooling crystallization (用冷却结晶的方法分离固体混合物及提纯盐类)



5.2 Examples of Applications of Simple Eutectic-System Phase Diagrams (简单低共熔系统相图应用举例)

4) Guidance for drug compatibility and pharmaceutical preparation: If the eutectic point of two solid drugs is near or below room temperature, they should not be blended in formulation to avoid forming pasty or liquid substances. A typical example is compound acetylsalicylic acid tablets.

4) 指导药物的配伍和制剂：两种固体药物的低共熔点如果接近室温或者室温以下，便不易混在一起配方， 以防形成糊状物或液态。如复方乙酰水杨酸片。

5.2 Examples of Applications of Simple Eutectic-System Phase Diagrams (简单低共熔系统相图应用举例)

5) Improving dosage forms to enhance therapeutic efficacy: Solid particles precipitated between the turning point and the plateau segment of a cooling curve are large and non-uniform, whereas the eutectic mixture precipitated at the eutectic point consists of fine and uniform microcrystals, as exemplified by fast-acting medicines.

5) 改良剂型增进药效：在步冷曲线转折点至水平段之间析出的固体颗粒大且不均匀，而在低共熔点析出的低共熔混合物则细小、均匀的微晶。如速效药。

5.3 Generate phase diagrams for formed compounds (生成化合物的相图)

Some two-component solid-liquid equilibrium systems may form compounds, giving rise to a third species. For example, $m_A + n_B = A_mB_n$. Although the number of species increases by one, there exists one independent chemical reaction R=1, so the number of components remains 2.

有些二组分固-液平衡系统可能生成化合物，产生第三个物种。例如$m_A + n_B = A_mB_n$；虽说物种数增加了1，但同时有一个独立的化学反应式R = 1，此时的组分数仍为2。

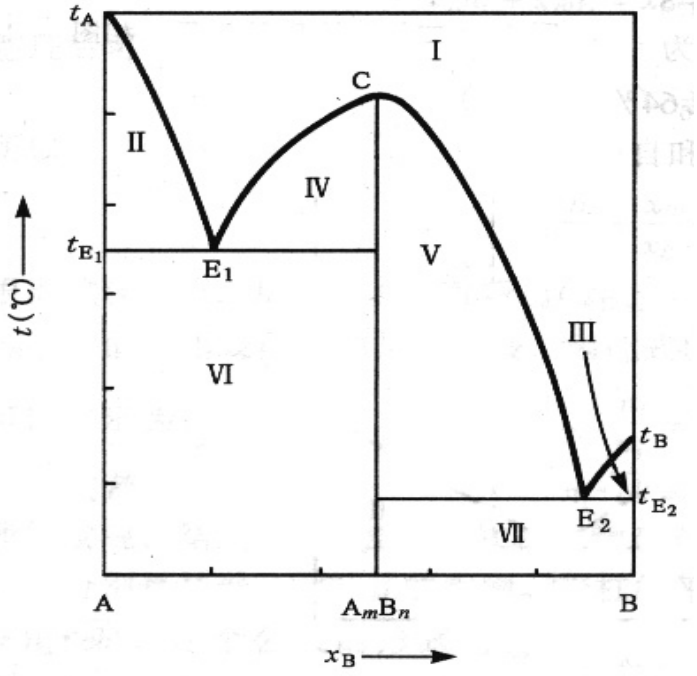
5.3 Generate phase diagrams for formed compounds (生成化合物的相图)

1) Phase diagrams for systems forming stable compounds

Stable compound: A compound that exists in both solid and liquid states with identical composition, also known as a congruent-melting-point compound.

1) 生成稳定化合物的相图

稳定化合物：该化合物在固、液态皆存在，且组成相同，又称为相合熔点化合物。



有稳定化合物生成的相图

5. Phase Diagrams of Binary Solid-Liquid Equilibrium Systems (二组分固-液平衡系统的相图)

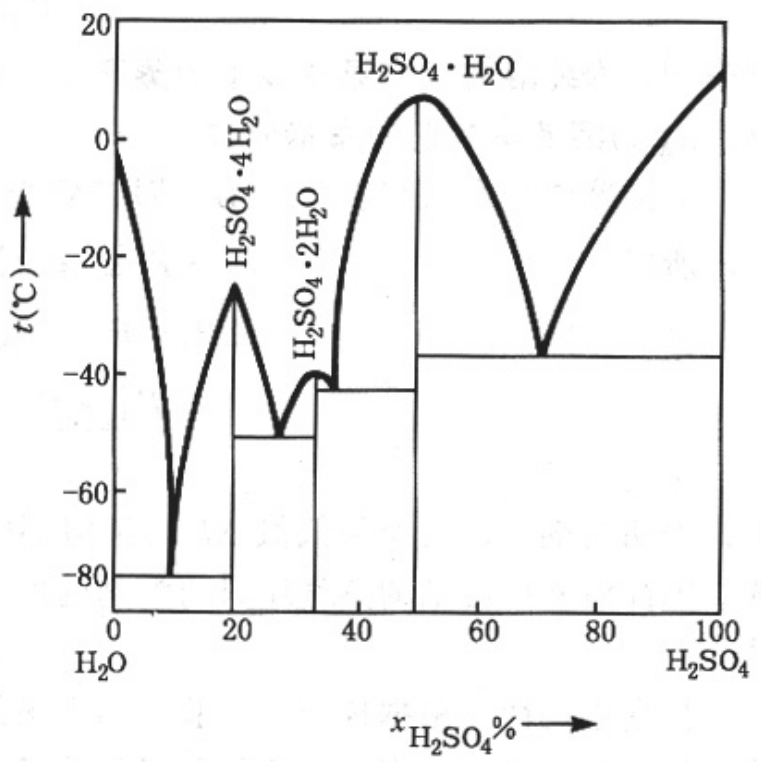
5.3 Generate phase diagrams for formed compounds (生成化合物的相图)

1) Phase diagrams for systems forming stable compounds

Some binary-component systems form multiple compounds. Examples of their phase diagrams are shown below.

1) 生成稳定化合物的相图

有的二组分系统形成多种化合物。其相图举例如下：



H_2SO_4-H_2O 的相图

5.3 Generate phase diagrams for formed compounds (生成化合物的相图)

2) Phase Diagrams Forming Unstable Compounds

Unstable compound: Upon heating, this compound decomposes below its melting point into another new solid phase and a melt with a composition different from that of the compound. It is also referred to as a compound with an incongruent melting point.

C_1(Compound, s) $\rightleftharpoons$ C_2 (s) + Melt

may be solid A, solid B, or a new compound. This reversible reaction is known as a peritectic reaction.

2) 生成不稳定化合物的相图

不稳定化合物：加热该化合物时，在熔点以下就分解为另一新固相及组成与化合物不同的熔液。又称为“不相合熔点”的化合物。

C_1(化合物，固) $\rightleftharpoons$ C_2 (固) + 熔液

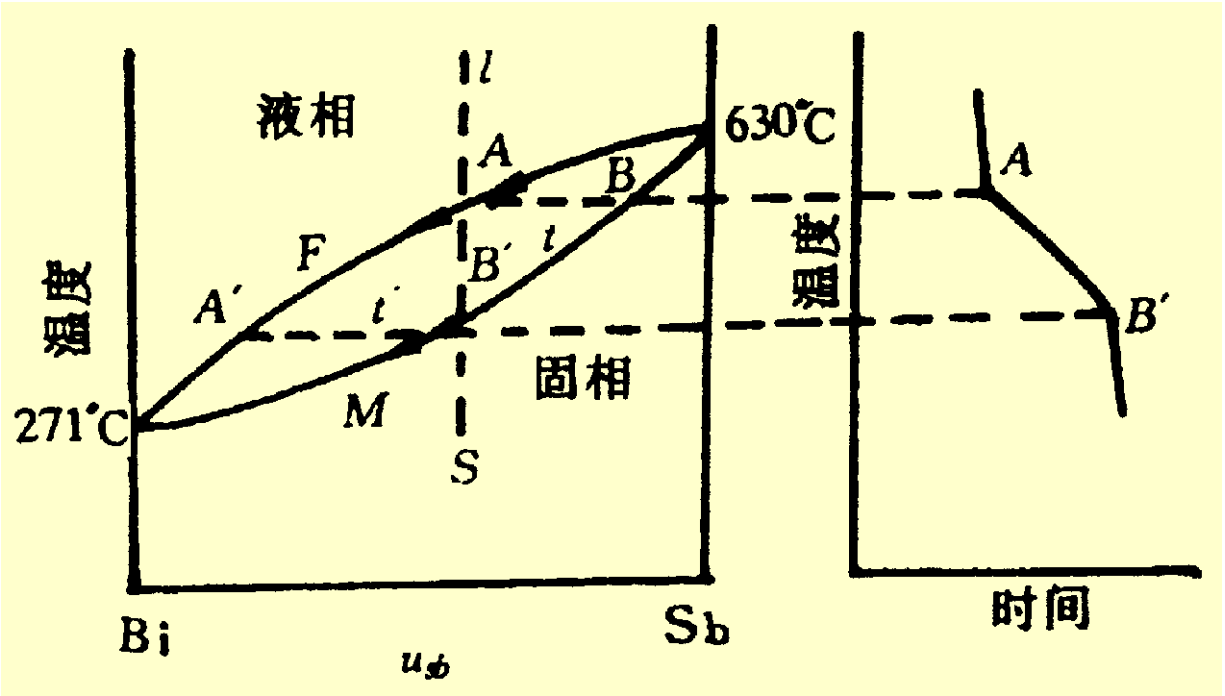
C_2(固)可以是A(固)或B(固)或是新化合物，该反应称为转熔反应，它是可逆的。

5. Phase Diagrams of Binary Solid-Liquid Equilibrium Systems (二组分固-液平衡系统的相图)

5.4 Phase diagrams with formation of solid mixtures (complete miscibility in the liquid phase, complete or partial miscibility in the solid phase) (有固态混合物生成的相图-液态完全互溶，固态完全互溶或部分互溶)

1) Phase diagram for a system with complete solid-phase miscibility

1) 固态完全互溶系统的相图



Bi-Sb系统的相图及步冷曲线（Phase Diagram and Cooling Curves of the Bi-Sb System)

5.4 Phase diagrams with formation of solid mixtures (complete miscibility in the liquid phase, complete or partial miscibility in the solid phase) (有固态混合物生成的相图-液态完全互溶，固态完全互溶或部分互溶)

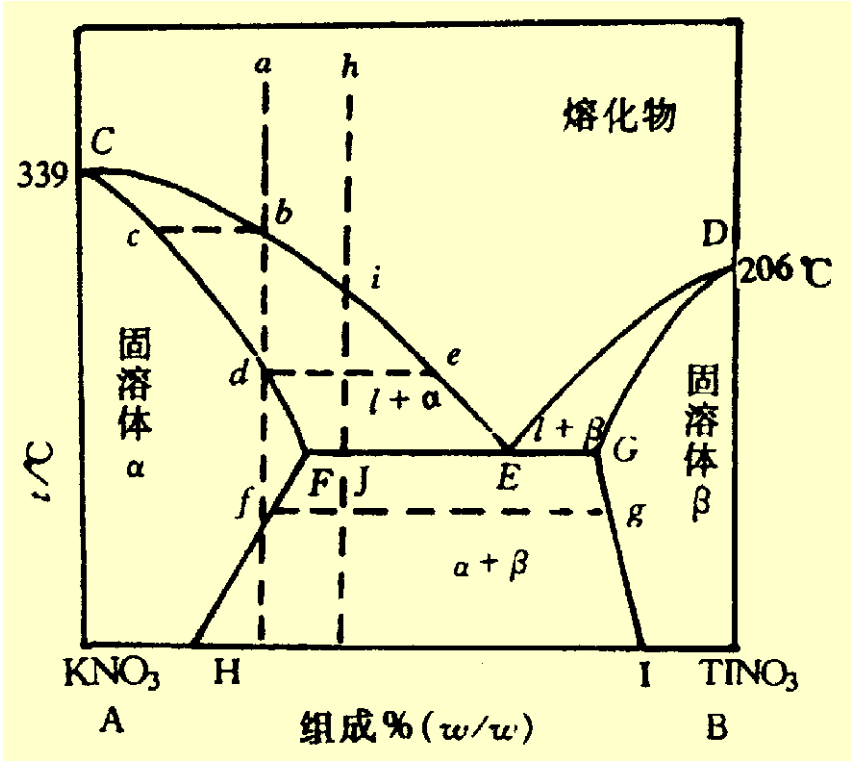
- Occasionally, phase diagrams for systems forming solid mixtures exhibit a maximum-melting-point or minimum-melting-point. At the maximum (or minimum) melting point, the liquid phase and solid phase have identical compositions, and a horizontal segment appears on the cooling curve.
- 有时在生成固态混合物的相图中出现最高熔点或最低熔点。在最高（或低）熔点处，液相与固相组成相同，步冷曲线上出现水平段。

5. Phase Diagrams of Binary Solid-Liquid Equilibrium Systems (二组分固-液平衡系统的相图)

5.4 Phase diagrams with formation of solid mixtures (complete miscibility in the liquid phase, complete or partial miscibility in the solid phase) (有固态混合物生成的相图-液态完全互溶，固态完全互溶或部分互溶)

2) Phase Diagram for a System with Partial Miscibility in the Solid Phase—Containing a Eutectic Point

2) 固态部分互溶系统的相图—有一个低共熔点

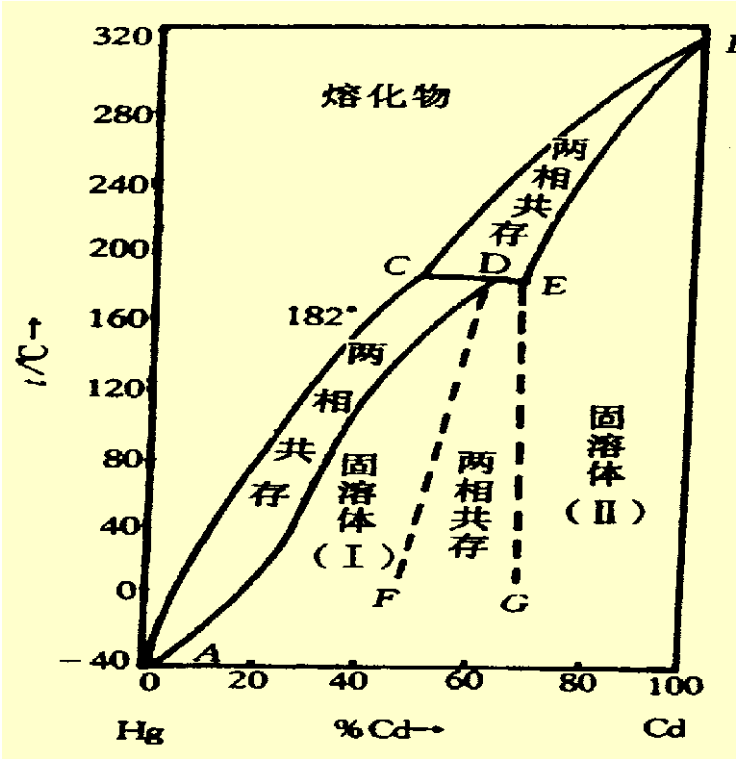


5. Phase Diagrams of Binary Solid-Liquid Equilibrium Systems (二组分固-液平衡系统的相图)

5.4 Phase diagrams with formation of solid mixtures (complete miscibility in the liquid phase, complete or partial miscibility in the solid phase) (有固态混合物生成的相图-液态完全互溶，固态完全互溶或部分互溶)

2) Phase Diagram for a System with Partial Miscibility in the Solid Phase—Containing a Eutectic Point

2) 固态部分互溶系统的相图—有一个转变温度



6. Phase Diagrams for Ternary Systems (三组分系统的相图)

- For a ternary system, according to the phase rule, f=3-Φ+2=5-Φ. When the degree of freedom reaches its minimum value of 0, the maximum number of phases is Φ=5. When the number of phases is at its minimum of 1, the maximum degree of freedom is f = 4. To obtain a two-dimensional phase diagram, pressure and temperature are kept constant; the maximum degree of freedom becomes 2 (corresponding to the compositions of two components).
- 对于三组分系统，由相律f=3-Φ+2=5-Φ；当自由度最少为0时，相数最大Φ=5；当相数最少为1时，自由度最大f=4；为得到平面图，将压力、温度保持恒定；f最多为2（两个组分的组成）。

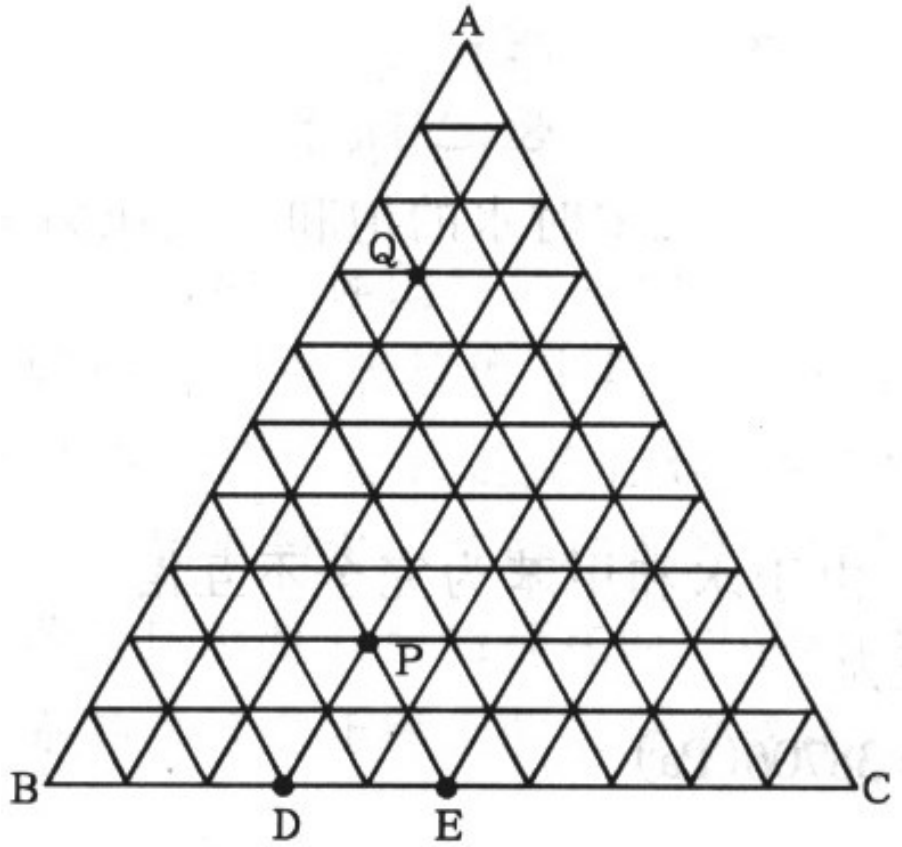
6. Phase Diagrams for Ternary Systems (三组分系统的相图)

6.1 Representation of Phase Diagrams for Ternary Equilibrium Systems(三组分平衡系统相图表示法)

Generally employs an equilateral triangle to denote the compositions of ternary systems, see Figure:

6.1 三组分平衡系统相图表示法

通常使用等边三角形来表示三组分系统的组成，见图：



Representation of Compositions for Ternary Systems

三组分系统组成表示法

6. Phase Diagrams for Ternary Systems (三组分系统的相图)

6.1 Representation of Phase Diagrams for Ternary Equilibrium Systems(三组分平衡系统相图表示法)

The characteristics of representing the compositions of ternary systems using triangular coordinates are as follows:

① Constant-content rule: At any point on a straight line parallel to one side of the triangle, the content of the component represented by the vertex opposite to this side remains identical.

② Constant-ratio rule: For ternary systems represented by all points on any straight line passing through a vertex of the triangle, the ratio between the contents of the other two components is constant.

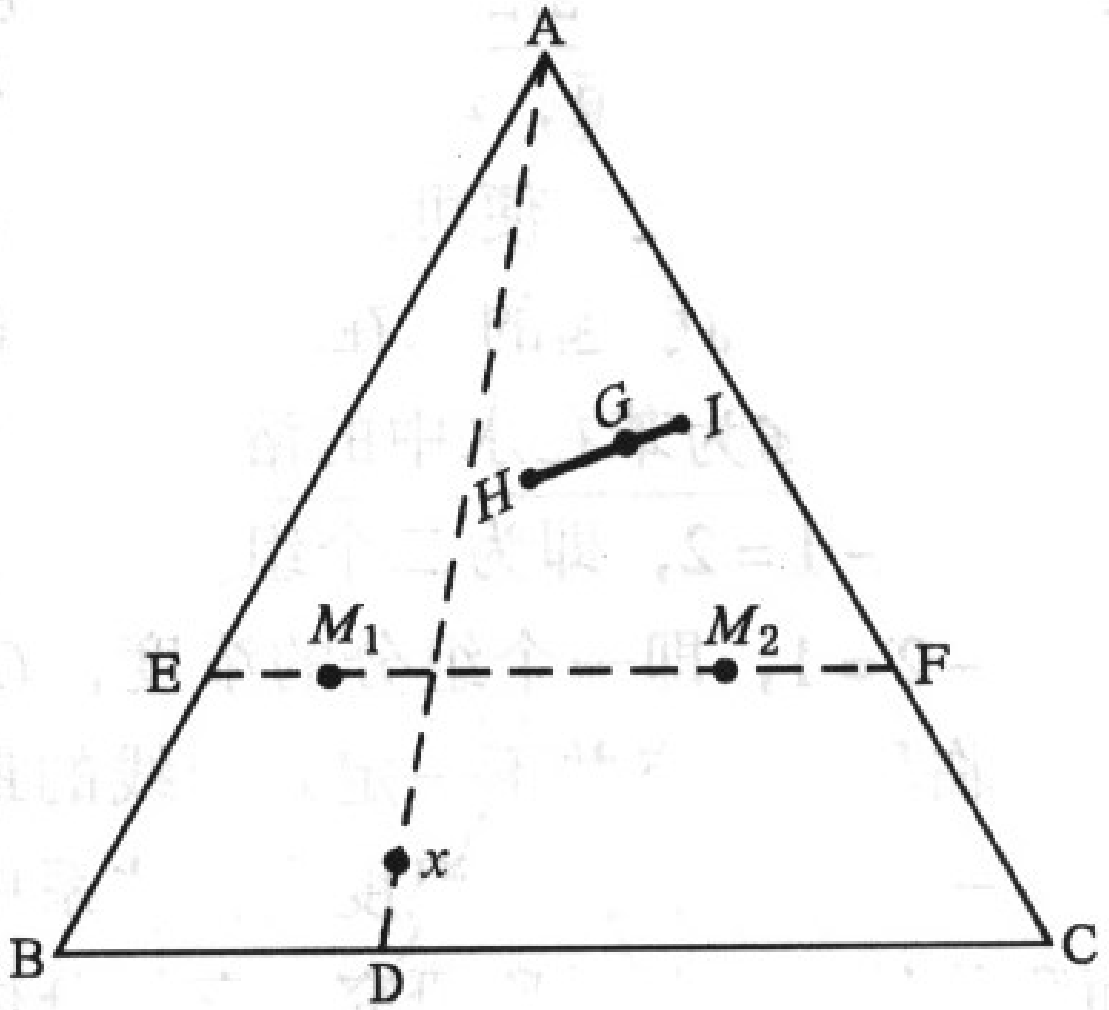
用三角形坐标表示三组分系统的组成有如下特点：

① 等含量规则：平行于三角形某边的直线上任意一点，含该边所对顶点代表组分的量都相同。

② 等比例规则：通过三角形某顶点的任一直线上各点代表的三组分系统中，另外两组分含量之比相同。

6. Phase Diagrams for Ternary Systems (三组分系统的相图)

6.1 Representation of Phase Diagrams for Ternary Equilibrium Systems(三组分平衡系统相图表示法)



6. Phase Diagrams for Ternary Systems (三组分系统的相图)

6.1 Representation of Phase Diagrams for Ternary Equilibrium Systems(三组分平衡系统相图表示法)

③ When two ternary systems with compositions H and I are mixed to form a new ternary system, the composition of the new system must lie on the connecting line HI. Its specific position is determined by the lever rule: that is $\frac{W_H}{W_I} = \frac{\overline{GI}}{\overline{GH}}$

If three ternary systems H, I and S are mixed, according to the lever rule, the system point of the resulting new ternary system is the centroid of triangle HIS. This is known as the centroid rule.

用三角形坐标表示三组分系统的组成有如下特点：

③ 若组成为H和I的两个三组分系统混合，形成新的三组分系统，其组成一定在HI的连线上，具体位置由杠杆规则确定：

即 $\frac{W_H}{W_I} = \frac{\overline{GI}}{\overline{GH}}$

若有三个三组分系统：H、I、S混合，由杠杆规则可知，新的三组分系统的物系点是三角形 HIS 的重心。称为重心规则。

6. Phase Diagrams for Ternary Systems (三组分系统的相图)

6.2 Ternary brine system (三组分盐水系统)

1) Systems with pure-salt solids

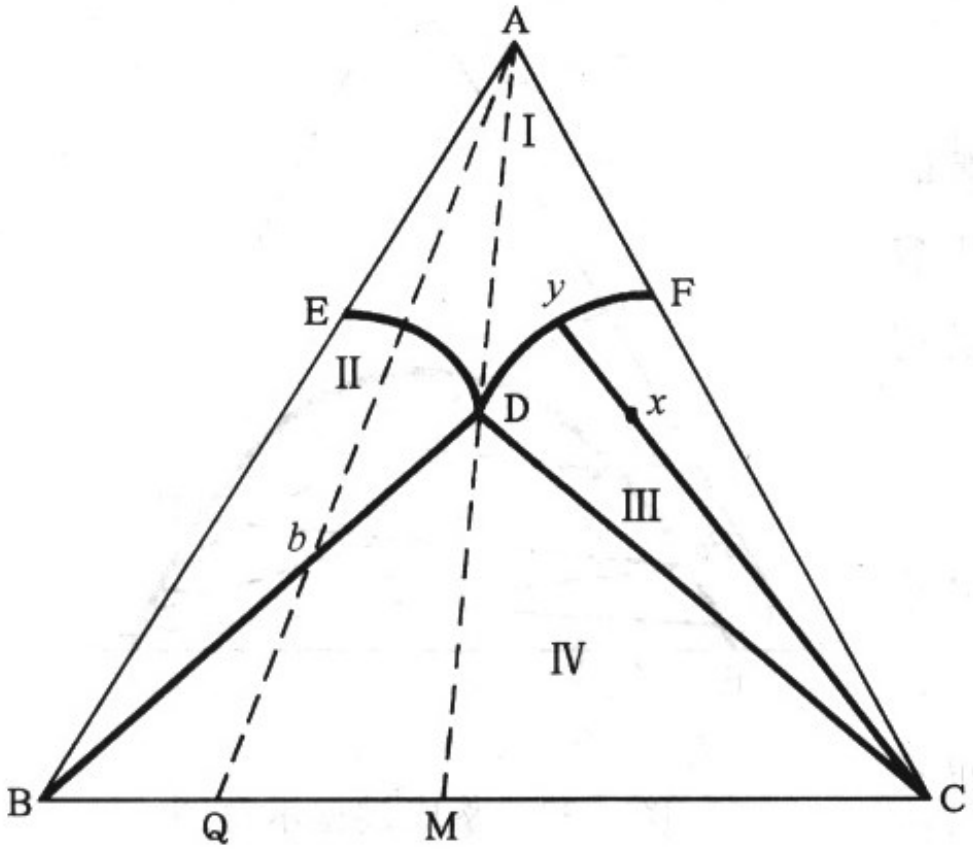
There are many types of ternary brine systems. Here we only discuss the phase diagrams of ternary systems containing two salts with a common ion. The common ion of the two salts can be either a cation or an anion.

1) 固体是纯盐的系统

三组分盐水系统种类很多，这里只讨论含有共同离子的两种盐的三组分系统相图。两种盐所含的共同离子可以是阳离子，也可以是阴离子。

6. Phase Diagrams for Ternary Systems (三组分系统的相图)

6.2 Ternary brine system (三组分盐水系统)



Phase Diagram of Ternary Brine System

三组分盐水系统相图

Applications of phase diagrams:

① Determine the possibility of obtaining a certain pure salt by adding water to a mixture of two salts.

② Determine the possibility of obtaining a certain pure salt by isothermal evaporation of a dilute solution containing two salts.

相图的应用：

① 判断在两种盐的混合物中加水， 获取某种纯盐的可能性。

② 判断两种盐的稀溶液等温蒸发， 获取某种纯盐的可能性。

6. Phase Diagrams for Ternary Systems (三组分系统的相图)

6.2 Ternary brine system (三组分盐水系统)

2) Hydrate-forming system

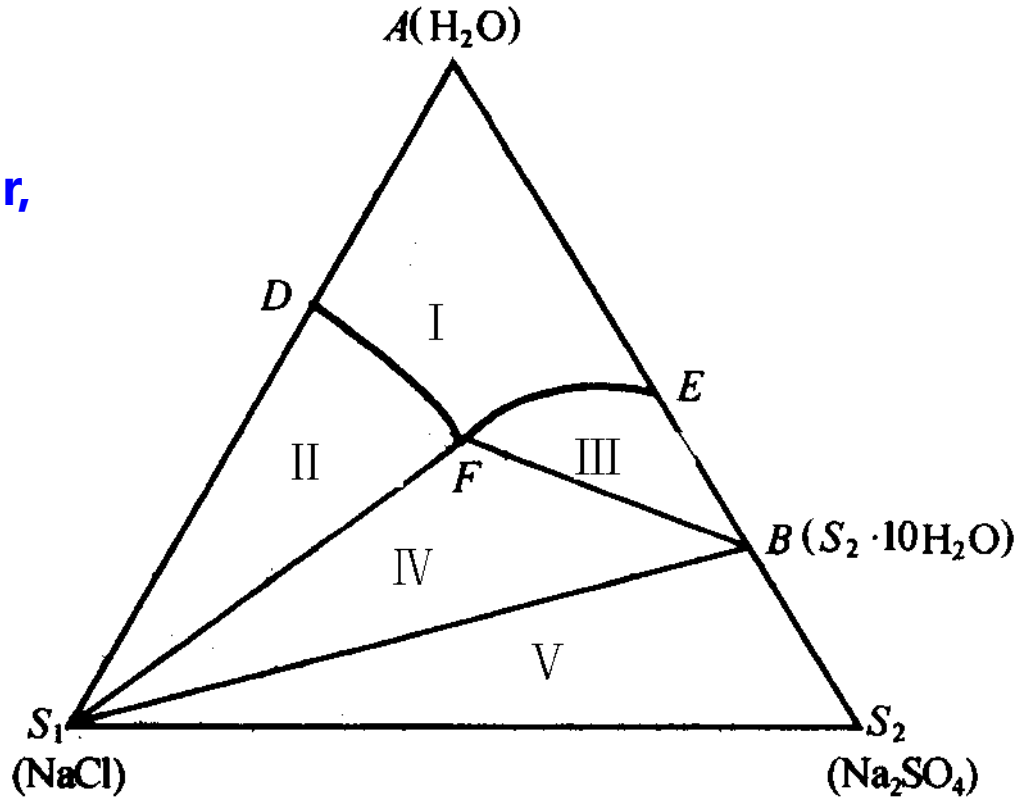
Take the H_2O-NaCl-Na_2SO_4 ternary system as an example.

Na_2SO_4 can form the solid hydrate $Na_2SO_4 \cdot 10H_2O$ with water, and its phase diagram is shown below:

2)生成水合物系统

以H_2O-NaCl-Na_2SO_4三组分系统为例，

Na_2SO_4与水能生成固态水合物$Na_2SO_4 \cdot 10H_2O$其相图为：



System with hydrate formation

有水合物生成的系统

6. Phase Diagrams for Ternary Systems (三组分系统的相图)

6.2 Ternary brine system (三组分盐水系统)

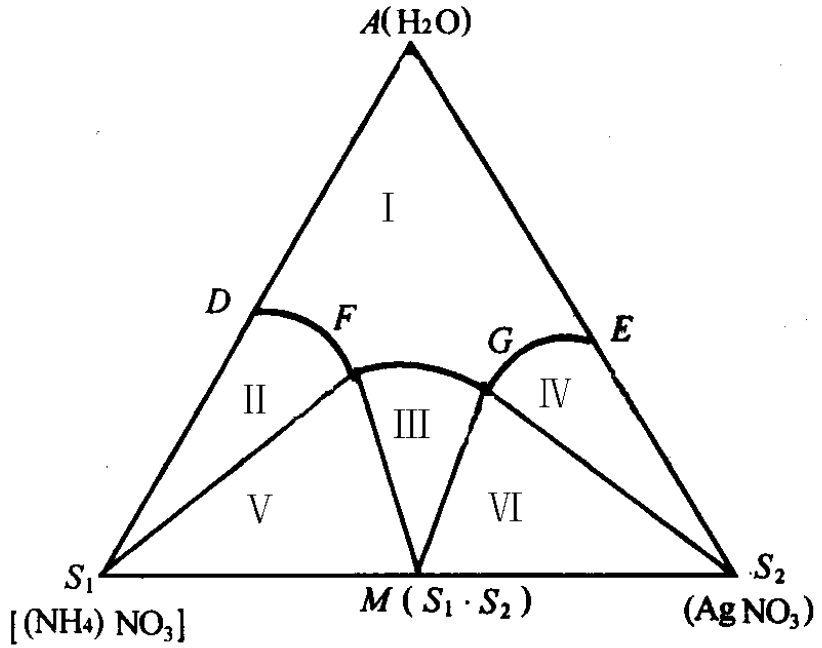
3) System with double-salt formation

① If a double salt can be formed by the two salts,

its phase diagram is shown below:

3) 生成复盐的系统

① 若两种盐能生成复盐，其相图为：



System with double-salt formation

生成复盐的系统

6. Phase Diagrams for Ternary Systems (三组分系统的相图)

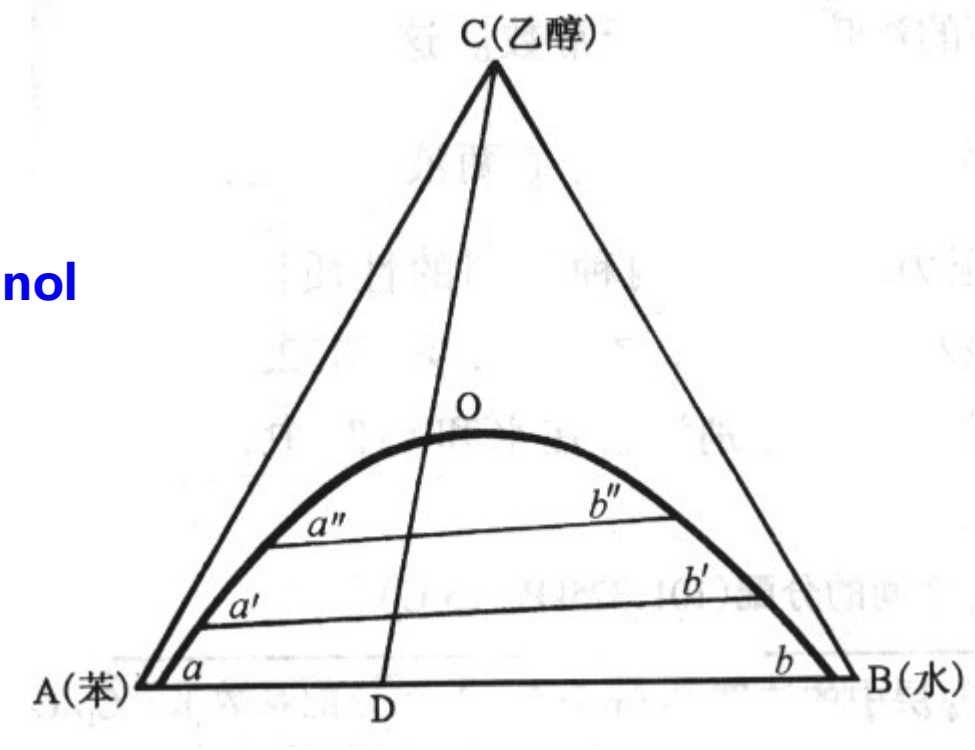
6.2 Ternary brine system (三组分盐水系统)

3) Ternary liquid systems with partial miscibility

① Phase diagram of a ternary liquid system with one pair of partially-miscible liquids, for example, the benzene-water-ethanol system. Its phase diagram is shown below:

3)部分互溶的三液系统

① 有一对部分互溶的三液系统相图，例如苯-水-乙醇系统。其相图如下:



Phase Diagram of Ternary System with One Pair of Partially Miscible Liquids

一对部分互溶的三组分系统相图

Chapter Summary and review (本章总结与复习)

1. Basic Concepts

① Phase: A phase is defined as a portion of a system with completely uniform physical and chemical properties. Distinct physical interfaces exist between different phases. ② Number of species: The number of chemical substances contained in an equilibrium system is called the number of species. ③ Number of independent components: The number of independent components, abbreviated as the number of components, refers to the minimum number of species sufficient to express the composition of all phases in a system. ④ Degrees of freedom: In an equilibrium system, the degrees of freedom are the maximum number of intensive properties that can be arbitrarily varied within a certain range without eliminating existing phases or generating new phases. ⑤ Phase rule: The phase rule describes the relationship among variables such as the number of phases, the number of independent components, and degrees of freedom (temperature, pressure, composition, etc.) in an equilibrium system. ⑥ Phase diagram: A phase diagram is a graph plotted from experimental data that illustrates the relationships between the phase states of a system and temperature, pressure, and composition. It cannot be derived from the phase rule and must be constructed based on experimental data.

1. 基本概念

① 相：系统中物理性质和化学性质完全均一的部分称为相。相与相之间有明显的物理界面

② 物种数：平衡系统中所含的化学物质数为物种数

③ 独立组分数：独立组分数简称组分数，为足以表示系统中所有各相组成所需的最少物种数

④ 自由度：平衡系统中，在不发生旧相消失或新相产生的条件下，在一定范围内可以任意改变的强度性质的最多数目为自由度

⑤ 相律：平衡系统中相数、独立组分数与自由度（温度、压力、组成……）等变量之间遵守的关系。

⑥ 相图：根据实验数据绘制的系统中相的状态与温度、压力、组成间相互关系的图形为相图。相图不能由相律绘制，只能根据一定的实验数据绘制。

Chapter Summary and review (本章总结与复习)

2. Basic Characteristics of One-Component Systems

The phase rule for a one-component system is: f= 1–φ+2=3-φ. A one-component system can be described by two variables, temperature and pressure, and a p-T phase diagram can be plotted.

① Phase diagram of water: It consists of three phase regions (gas, liquid and solid phases, f=2, three two-phase equilibrium lines (gas-liquid, gas-solid and solid-liquid equilibrium, f=1, and one triple point (coexistence of gas-liquid-solid phases, f=0. The triple-point temperature of water is 273.16 K, and the pressure is 0.610 6 kPa.

② Clausius-Clapeyron equation: The quantitative relationship between T and p for two-phase equilibrium in a one-component system is described by the Clapeyron equation. For gas-liquid or gas-solid equilibrium of pure substances involving the gas phase, the Clausius-Clapeyron equation can be applied.

2. 单组分系统的基本特征

单组分系统的相律为： f= 1–φ+2=3-φ。单组分系统可用温度和压力两个变量描述，绘制 p-T相图。

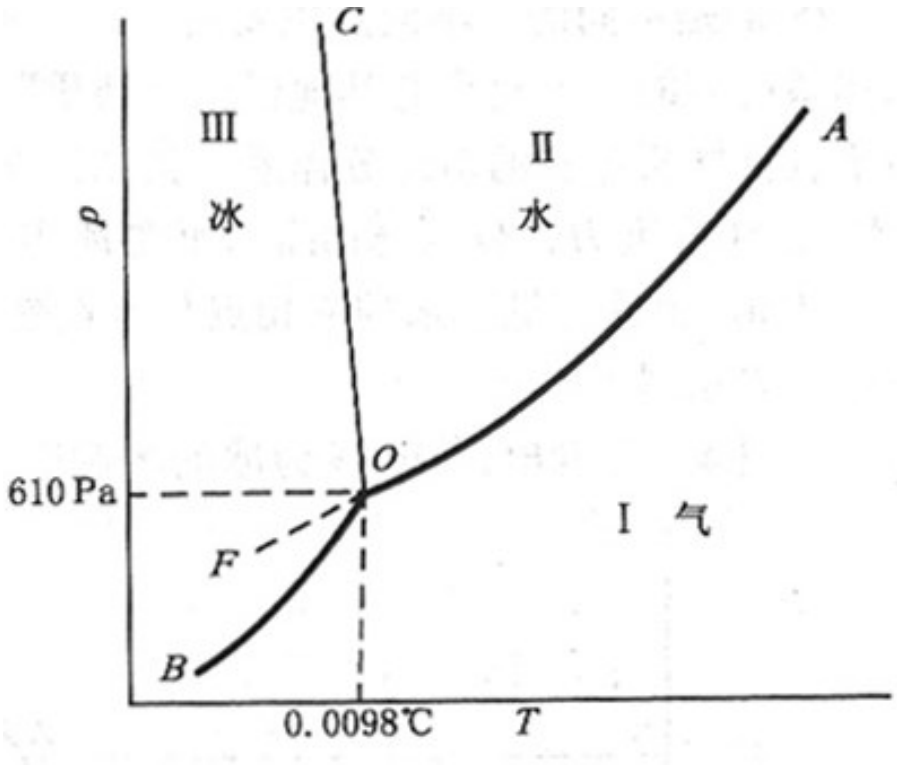
① 水的相图：有三个相区（气相、液相、固相，f=2）、三条两相平衡线（气-液、气-固和固-液两相平衡，f=1）和一个三相点（气-液-固三相共存，f=0）；水的三相点温度为 273.16K，压力为 0.610 6 kPa。

② 克劳修斯-克拉珀龙方程：单组分系统两相平衡共存时T与p的定量关系式可由克拉珀龙方程式描述。对于有气相参与的纯物质气-液两相或气-固两相平衡，可用克劳修斯-克拉珀龙方程描述。

Chapter Summary and review (本章总结与复习)

2. Basic Characteristics of One-Component Systems

2. 单组分系统的基本特征



Phase Diagram of Water

水的相图

Regions: single-phase regions, f=2

Lines: two-phase equilibrium lines, f=1, the relationship between P and T obeys the Clapeyron equation.

Points: triple point, f=0

面：单相区， f = 2

线：两相平衡线，f = 1，P与T的关系服从克拉贝龙方程

点：三相点，f = 0

Chapter Summary and review (本章总结与复习)

3. Characteristics of Binary-Component System Phase Diagrams

① The curves represent two-phase equilibrium lines, and the points on the curves are phase points.

② The horizontal lines in the diagram are three-phase lines. The three phase points are located at the two ends and the intersection of the horizontal line, and f=0 on the three-phase line.

③ The vertical lines in the diagram denote single-component compounds. For a stable compound, the top of the vertical line intersects with a curve; for an unstable compound, the top of the vertical line intersects with a horizontal line, forming a T-shape.

④ The lever rule is applicable within regions of two-phase coexistence.

⑤ f=2 in single-phase regions, f=1 in two-phase regions, and f=0 on three-phase lines.

3. 二组分系统相图的特征

① 曲线为两相平衡线，曲线上的点为相点；

② 图中水平线为三相线，三个相点分为在水平线的两端和交点上，三相线上 f= 0；

③ 图中垂直线表示单组分化合物。若为稳定化合物，垂直线顶端与曲线相交，若为不稳定化合物，垂直线顶端与水平线相交，呈T字型；

④ 两相平衡共存区内可适用杠杆规则；

⑤ 单相区f=2，两相区f=1，三相线f= 0。

Chapter Summary and review (本章总结与复习)

4. Characteristics of Ternary-Component System Phase Diagrams

① The fan-shaped regions and cap-shaped regions in the phase diagram are two-phase coexistence regions. The line segments within these regions are tie lines, to which the lever rule applies, and f=1.

② Triangular regions represent three-phase coexistence regions. The composition and state of each phase are described by the vertices of the triangle, and f=0.

③ The center-of-gravity rule is applicable to three-phase coexistence regions.

4. 三组分系统相图的特征

① 相图中的扇形区和帽形区是两相共存区， 区域中的线段为结线，适用杠杆规则，f=1；

② 三角形区为三相共存区，每相的成分和状态由三角形的顶点来描述，f=0 ；

③ 三相共存区可适用重心规则。

谢谢各位同学！